

Time Series and Stochastic Processes exams

Angry Teachers, Folklore

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Description

See updates at https://github.com/bdemeshev/tssp_exams.

Many more problems can be found at https://github.com/bdemeshev/stochastic_pro.

Click on red hyperlinks inside pdf, you can get to the answers and back!

Any comments? Bugs? https://github.com/bdemeshev/tssp_hse_exams/issues/.

The order of topics has changed substantially after the first course iteration in 2020-21. The interested reader may find relevant exercises by looking through all 2020-21 exams.

Greetings to the contributors

Here we describe only the style guidelines and typical errors. For more information on tex one may read the [book](#) by K. Vorontsov.

1. Use decimal point as a separator: 3.14 — good style, 3,14 — bad style. This goes against russian tradition, but favors copy-pasting numbers in software for computations.
2. Use `\[...\]` for display math formulas. Do not use `$$...$$`!
3. Use `cases` for systems of equations, `align*` for multiline formulas, `enumerate` for enumerations.
4. Inside formulas use `\text{...}` to write text.
5. Use `\ldots` for ellipsis.
6. You can find useful macros in the preamble, like `\P`, `\E`, `\Var`, `\Cov`, `\Corr`, `\cN`.
7. Use backslash before functions: `\ln`, `\exp`, `\cos...`
8. Use booktabs style for tables. You may use online [tablesgenerator](#). Choose booktabs table style instead of default table style.
9. Respect the letter ë! :)
10. Start every sentence in tex source from a new line. There will be no additional newlines in final pdf but tex file will be easier to read.
11. For multiplication use `\cdot`. Please never use `*` :)

1 October exam

1.1 2025-2026

FINAL SALE: Every problem will bring you 10 99.99 points!

Scratch paper price: ~~10 points per sheet only~~ **FREE**¹!

1. Dragon Erik has three towns to kidnap a princess from: A , B and C . He kidnaps the first princess from town A and chooses every next town according to a Markov chain with transition matrix T , where states are written in alphabetical order:

$$T = \begin{pmatrix} 0.1 & 0.2 & ? \\ ? & 0 & 0.7 \\ 0.2 & ? & 0.5 \end{pmatrix}.$$

- (a) [2] Fill in the missing values.
 - (b) [2] Draw the graph of the Markov chain.
 - (c) [3] What is the probability that the third princess will be from town B ?
 - (d) [3] What is the probability that the second princess was from A given that the third princess was from B ?
2. Every minute the cat Tikhon says «meow» with probability $1/3$ or «purr» with probability $2/3$ independently of all other words. For every «purr» that follows «meow» you get 1 PEPE, for every «meow» that follows «purr» you pay 3 PEPEs. For «purr» that follows «purr» or «meow» that follows «meow» you get nothing.

You have just earned 1 PEPE. Let T be the time to get the next positive reward.

- (a) [5] Find $\mathbb{E}(T)$.
 - (b) [5] Find $\text{Var}(T)$.
3. [10] Gumbatali starts with zero initial sum $S_0 = 0$. Every minute he either wins $X_t = 1$ dollar with probability $1/3$ or pays 1 dollar, $X_t = -1$, with probability $2/3$. He can't go neither below $S_t = -2$ nor above $S_t = 3$, so that

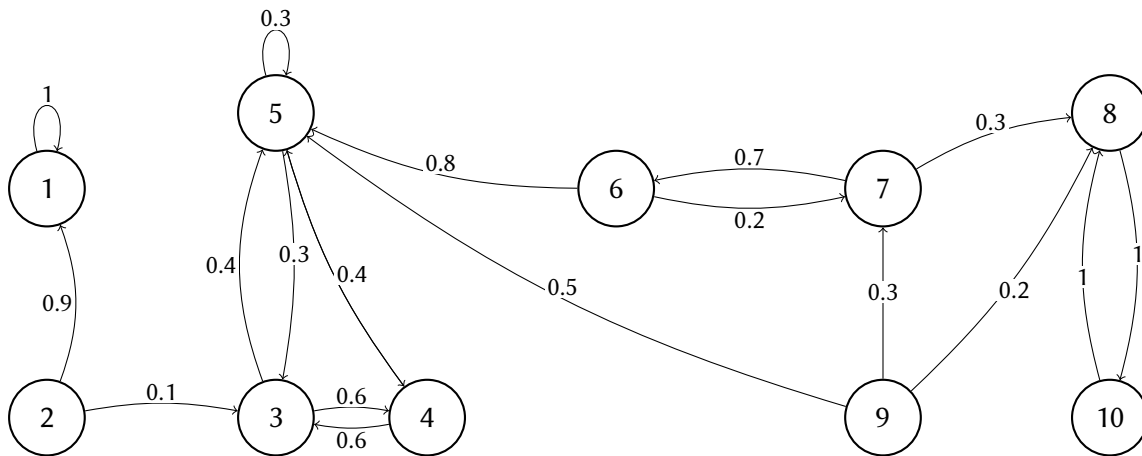
$$S_t = \min\{3, \max\{-2, S_{t-1} + X_t\}\}.$$

The random variables (X_t) are independent.

Find the stationary distribution of S_t .

¹Members only deal!

4. Consider the Markov chain:



- [5] Using a chainsaw cut the Markov chain into communicating classes and classify them as transient, positive recurrent or null-recurrent.
- [1] Is the chain irreducible?
- [4] Find the period of every state.

5. Let X be a Poisson distributed random variable with intensity $\lambda = 2$. The probability mass function of a Poisson random variable is:

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- [4] Derive the moment generating function $M_X(t) = \mathbb{E}(e^{tX})$.
 - [4] Using moment generating function find the mean and variance of X .
 - [2] Let $X_1 \sim \text{Pois}(\lambda = 2)$ and $X_2 \sim \text{Pois}(\lambda = 3)$ be independent random variables. Find the moment generating function of $S = X_1 + X_2$.
6. By the mayor's decree, the city installs T_n New Year's trees each day, $n = 1, 2, \dots$, starting from October 20th. The random variables $T_1, T_2, \dots$ are independent and take any natural value from 4 to 10 with equal probabilities. The amounts of snow (W_n) are independent and uniformly distributed on $[0; 1/n]$.

Determine the probability limits for the following sequences:

- [2] mayor's New Year kpi, $X_n = \frac{1}{n} \sum_{i=1}^n T_i$
- [3] the New Year's mood given by $Y_n = \frac{2X_n+1}{X_n+3}$
- [5] The average number of snowmen that can be built each day: $S_n = \frac{1}{n} \sum_{i=1}^n W_i$.

1.2 2024-2025

- [10] Michael stands in a corner of a hexagonal room with white soft walls. Stochastic Processes course has caused him a deep trauma. Michael has insomnia and two independent personal identities. At each iteration each personal identity moves from one vertex of the hexagon to the adjacent one. Two personal identities move independently with equal probabilities in both directions.

Consider the Markov chain where the state is a relative position of two identities of Michael in a hexagon.

- [4] Draw the diagram of chain states and find the transition matrix.
 - [2] Classify the states as transient or recurrent.
 - [4] Which proportion of his eternal life Michael spends in a perfect harmony with himself (two personal identities are located at the same vertex)?
- [10] Compare sigma-algebras and conditional expected values!

- (a) [5] Consider sigma-algebras

$$\mathcal{F}_1 = \sigma(X), \quad \mathcal{F}_2 = \sigma(Y), \quad \mathcal{F}_3 = \sigma(X, Y), \quad \mathcal{F}_4 = \sigma(X + Y, X - Y).$$

Which of them are always equal? Which sigma-algebra is always a subset of another one?

- (b) [5] Consider random variables

$$R_1 = \mathbb{E}(X | Y), \quad R_2 = \mathbb{E}(1/X | Y), \quad R_3 = \mathbb{E}(1/X | 1/Y), \quad R_4 = \mathbb{E}(X | 1/Y).$$

Which of them are always equal provided that they exist?

3. [10] The random variables (X_n) are independent and uniform on $[0; 1]$ and $S_n = X_1 + X_2 + \cdots + X_n$.

- (a) [4] Find the moment generating function of X_1 .
(b) [2] Find the moment generating function of $R = S_{10} - 5$.
(c) [4] Find $\mathbb{E}(X_1 | S_3)$.

4. [10] I throw a fair coin. Let Y be the number of throws until I obtain the sequence head-tail-head.
- (a) [4] Find $\mathbb{E}(Y)$.
 - (b) [6] Find $\mathbb{E}(Y^2)$ and hence $\text{Var}(Y)$.
5. [10] Consider two non-random sequences, $h_n = 1/n$ and $t_n = 1 + 1/n$. Elon Musk throws a fair coin once and selects the sequence (h_n) if it lands on head and selects the sequence (t_n) otherwise. Hence Elon obtains a random sequence (X_n) .
- (a) [2] What is the distribution of X_5 ?
 - (b) [4] What is the distribution of $\lim X_n$?
 - (c) [4] Write $\text{plim } X_n$ explicitly as a function of X_1 .
6. [10] The random variables (X_n) are independent and they have Poisson distribution with rate $\lambda = 2$. Consider the cumulative sum $S_n = X_1 + X_2 + \dots + X_n$ with $S_0 = 0$ and the natural filtration $\mathcal{F}_n = \sigma(X_1, X_2, \dots, X_n)$.
- (a) [4] Provide two examples of events that belong to $\mathcal{F}_9$ but do not belong to $\mathcal{F}_7$.
 - (b) [6] Find all constants a and b such that $M_n = S_n + a + b \cdot n$ is a martingale.

Hint: if $R \sim \text{Pois}(\lambda)$ then $\mathbb{E}(R) = \lambda$ and $\text{Var}(R) = \lambda$.

1.3 2023-2024

Short rules: 90 minutes, offline. You may use one A4 cheat sheet.

Date: 2023-10-21.

1. The hedgehog Melissa starts at the vertex A of a triangle $\triangle ABC$. Each minute she randomly moves to an adjacent vertex with probabilities $\mathbb{P}(A \rightarrow B) = 0.7$, $\mathbb{P}(A \rightarrow C) = 0.3$, $\mathbb{P}(B \rightarrow C) = \mathbb{P}(B \rightarrow A) = 0.5$, $\mathbb{P}(C \rightarrow B) = \mathbb{P}(C \rightarrow A) = 0.5$.
- (a) What is the probability that she will be in vertex B after 3 steps?
 - (b) Write down the transition matrix of this Markov chain.
 - (c) What proportion of time Melissa will spend in each state in the long run?
2. The number of players N who will win the lottery is a random variable with probability mass function $\mathbb{P}(N = k) = 7 \cdot 0.3^k / 3$ for $k \geq 1$. Each player will get a random prize $X_i \sim U[0; 1]$. All random variables are independent. Let S be the sum of all the prizes.
- (a) Find $\mathbb{E}(S | N)$ and conditional moment generating function $M_{S|N}(u)$.
 - (b) Find the unconditional moment generating function $M_S(u)$.
 - (c) What is the probabilistic meaning of $M_S''(0) - (M_S'(0))^2$?

Note: you don't need to calculate the value in (c).

3. Consider the stochastic process (X_n) , where X_0 is uniform on $[0; 2]$ and $X_n = (1 + X_{n-1})/2$.
- (a) Find $\mathbb{E}(X_n)$ and $\text{Var}(X_n)$.
 - (b) Find the probability limit $\text{plim } X_n$.
4. Students arrive in the Grusha café according to the Poisson arrival process (X_t) with constant rate λ . The probability of no visitors during 5 minutes is 0.05.
- (a) Find the value of λ .
 - (b) Find the variance and expected number of arrivals between 5 pm and 8 pm.

- (c) What is the probability of exactly 5 arrivals between 5 pm and 8 pm?
5. The random variables X_1 and X_2 are independent and normally distributed, $X_1 \sim \mathcal{N}(1; 1)$, $X_2 \sim \mathcal{N}(2; 2)$. I choose X_1 with probability 0.3 and X_2 with probability 0.7 without knowing their values. Casino pays me the value Y that is equal to the chosen random variable. Let the indicator I be equal to 1 if I choose X_1 and 0 otherwise.

- (a) Express Y in terms of X_1 , X_2 and I .
- (b) Find $\mathbb{E}(Y | I)$, $\text{Var}(Y | I)$.
- (c) Find $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

6. The joint distribution of X and Y is given in the table

	$X = -2$	$X = 0$	$X = 2$
$Y = -1$	0.1	0.2	0.3
$Y = 1$	0.2	0.1	0.1

- (a) Explicitly find the σ -algebra $\sigma(X)$.
- (b) How many elements are there in $\sigma(X \cdot Y)$?

1.4 2022-2023

Short rules: 120 minutes, online and offline. You may use one A4 cheat sheet.

Date: 2022-10-29

1. [10] The random variables X_i are independent and uniformly distributed on $[0; 2]$. Find

$$\lim_{n \rightarrow \infty} \frac{(X_1 - \bar{X})^3 + (X_2 - \bar{X})^3 + \dots + (X_n - \bar{X})^3}{n + 2022}.$$

2. A Hedgehog starts at the point $x = 2$ on the real line. Every minute he moves one step left with probability 0.3 or one step right with probability 0.7. There are two exceptions from this rule: the absorbing point $x = 0$ and the reflecting barrier at $x = 3$.

If the Hedgehog reaches the absorbing point $x = 0$ then he stops moving and stays there. If the Hedgehog reaches the reflecting barrier $x = 3$ then his next move will be one step left with probability 1.

- (a) [2] Write the transition matrix of this Markov chain.
- (b) [3] What is the probability that Hedgehog will be at $x = 1$ after exactly 3 steps?
- (c) [5] What is the expected time to reach the absorbing state?

3. The random variables X_i are independent and they take values $+1$ or -1 with equal probability.

- (a) [3] Explicitly list all the events in sigma-algebra $\sigma(X_1 \cdot X_2)$.
- (b) [3] Pavel says that he knows only whether X_1 and X_3 are equal. How will you describe his knowledge with sigma-algebra?
- (c) [4] How many events are in the sigma-algebra $\sigma(X_1, X_1 + X_2, X_1 + X_2 + X_3)$?

4. Masha receives on average 10 sms per minute. Sms arrival is well described by the Poisson process.

- (a) [3] What is the probability that Masha receives exactly 10 sms in the next 40 seconds?
- (b) [3] Masha just received an sms. What is the probability that she will wait more than 2.5 seconds before the next one?

- (c) [4] Find the covariance between the number of sms in the first 3 minutes and the number of sms in the first 10 minutes.
5. The random variables X_i are independent and they take values $+1$ or -1 with equal probability.
- (a) [3] Find $\mathbb{E}(X_3 \mid X_1, X_2)$, $\mathbb{E}(X_3 \mid X_1 + X_3)$.
- (b) [3] Find $\text{Var}(X_3 \mid X_1, X_2, X_3)$, $\text{Var}(X_3 \mid X_1 + X_3)$.
- (c) [4] Let Y_n be equal to $\mathbb{E}(X_1 + \dots + X_{2022} \mid X_1, X_2, \dots, X_n)$.
Is the process $Y_1, Y_2, \dots, Y_{2022}$ a martingale?
6. Consider a Wiener process (W_t) .
- (a) [4] Let $Y_t = tW_{2t}$. What is the distribution of $Y_t - Y_s$ for $t \geq s$? Is Y_t a Wiener process?
- (b) [6] Find a constant α such that $M_t = W_t^3 + \alpha t W_t$ is a martingale.

1.5 2021-2022

Short rules: 120 minutes, online without proctoring. You may use any source you want but don't cheat.
Date: 2021-10-28

1. (10 points) Consider the Markov chain with the transition matrix

$$P = \begin{pmatrix} 0.2 & 0.2 & 0 & 0.6 \\ 0.3 & 0.3 & 0.4 & 0 \\ 0 & 0 & 0.1 & 0.9 \\ 0 & 0 & 0.8 & 0.2 \end{pmatrix}.$$

- (a) (3 points) Split the chain in classes and classify them into closed or not closed.
- (b) (2 points) Classify the states into recurrent or transient.
- (c) (5 points) A Hedgehog starts in the state one and moves randomly between states according to the transition matrix.
What is the approximate probability that the Hedgehog will be in the state four after 10^{2021} moves?

Note: state number is the row (or column) number.

2. (10 points) Gleb Zheglov catches one criminal every day. With probability 0.2 the caught criminal is replaced by w new criminals. Initially there are n criminals in the town.

What is the expected time to the ultimate crime eradication in the town?

- (a) (4 points) Solve the problem for $w = 1$ and $n = 1$.
- (b) (6 points) Solve the problem for arbitrary w and n .
3. (10 points) The random variables X_i are independent and uniformly distributed on $[0; 1]$. Find the probability limit

$$\text{plim}_{n \rightarrow \infty} \max \left\{ \frac{\sum_{i=1}^n X_i}{n}, \frac{2 \sum_{i=1}^n X_i^2}{n} \right\}.$$

4. (10 points) Taxis arrive to the station according to the Poisson process with rate 1 per 5 minutes.

Let Y_t be the number of taxis that will arrive between 0 and t minutes.

- (a) (2 points) Sketch the expected value of Y_t as a function of t .
- (b) (8 points) Sketch the probability $\mathbb{P}(Y_t = Y_{60})$ as a function of t .

Note: special points like intercepts or extrema should be explicitly marked.

5. (10 points) Prince Myshkin throws a fair coin until two consecutive heads appear. Let N be the number of throws.

Find the moment generating function of N .

Hint: you may use the first step approach.

6. (20 points) Vincenzo Peruggia makes attempts to steal the Mona Lisa painting until the first success. Each attempt is successful with probability 0.1.

Let X be the number of attempts and $Z = \min\{X, 5\}$.

- (a) (5 points) How many events are in sigma-algebras $\sigma(Z)$ and $\sigma(X)$?
- (b) (5 points) If possible provide an example of events A and B such that: $A \in \sigma(Z)$ but $A \notin \sigma(X)$; $B \in \sigma(X)$ but $B \notin \sigma(Z)$.
- (c) (10 points) Find $\mathbb{E}(Z | X)$ and $\mathbb{E}(X | Z)$.

1.6 2021-2022 retake

Short rules: 120 minutes, online without proctoring. You may use any source you want but don't cheat.

1. (10 points) Consider the Markov chain with the transition matrix

$$P = \begin{pmatrix} 0.2 & 0.2 & 0 & 0.6 & 0 \\ 0.3 & 0.3 & 0.4 & 0 & 0 \\ 0 & 0 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0.8 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) (3 points) Split the chain in classes and classify them into closed or not closed.
- (b) (2 points) Classify the states into recurrent or transient.
- (c) (5 points) A Hedgehog starts in the state one and moves randomly between states according to the transition matrix.

What is the approximate probability that the Hedgehog will be in the state four after 10^{2021} moves?

Note: state number is the row (or column) number.

2. (10 points) Consider infinite ladder with steps numbered from 0 to infinity. I start at step 0. Every day with probability u I go one step up. With probability d I go one step down. With probability $1 - u - d$ I stay on the same step.

If I am at step 0 then I stay there with probability $1 - u$ because it's impossible to go down.

Consider the case $d > u$.

What is the probability that I will be at step 0 after 10^{1000} days?

3. (10 points) The random variables X_i are independent and uniformly distributed on $[0; 2]$. Find the probability limit

$$\text{plim}_{n \rightarrow \infty} \max \left\{ \frac{\sum_{i=1}^{10} X_i}{n}, \frac{\sum_{i=1}^n X_i^3}{n+1} \right\}.$$

4. (10 points) Taxis arrive to the station according to the Poisson process with rate 1 per 5 minutes.

Let Y_t be the number of taxis that will arrive between 0 and t minutes.

- (a) (5 points) Sketch the probability $\mathbb{P}(Y_{t+3} = 1 | Y_t = 0)$ as a function of t .
- (b) (5 points) Sketch the covariance $\text{Cov}(Y_t, Y_{60})$ as a function of t .

Note: special points like intercepts or extrema should be explicitly marked.

5. (10 points) The moment generating function of a random variable X is $1/(1 - 2t)$.
- Find the moment generating function of $2X$.
 - Find the moment generating function of $X + Y$ where X and Y are independent and identically distributed.
 - Do you remember the sum of geometric progression? Find $\mathbb{E}(X^{2021})$.
6. (20 points) Variables $X_1, X_2, \dots, X_{100}$ are independent and identically distributed with mean 1 and variance 2. Each X_i has only three possible values: 0, 1, and 2.
- (5 points) How many events are in sigma-algebras $\sigma(X_1, X_2)$ and $\sigma(X_1 - X_2)$?
 - (5 points) If possible provide an example of events A and B such that: $A \in \sigma(X_1, X_2)$ but $A \notin \sigma(X_1 - X_2)$; $B \in \sigma(X_1 - X_2)$ but $B \notin \sigma(X_1, X_2)$.
 - (10 points) Find $\mathbb{E}(X_1 + \dots + X_{100} \mid X_1 + \dots + X_{50})$ and $\mathbb{E}(X_1 + \dots + X_{50} \mid X_1 + \dots + X_{100})$.

1.7 2020-2021

Here (W_t) denotes the standard Wiener process.

Date: 2020-10-30

1. For $r < s < t < u$ find the following expected values

- $\mathbb{E}((W_u - W_t)^2(W_s - W_r)^2)$;
- $\mathbb{E}((W_u - W_s)(W_t - W_r))$;
- $\mathbb{E}((W_t - W_r)(W_s - W_r)^2)$;
- $\mathbb{E}(W_r W_s W_t)$;
- $\mathbb{E}(W_r W_s W_t \mid W_s)$;

2. Consider Ito process X_t

$$dX_t = \exp(t)W_t dt + \exp(2W_t) dW_t, \quad X_0 = 1.$$

Consider two processes, $A_t = 1 + t^2 + X_t^3$ and $B_t = 1 + t^2 + X_t^3 W_t^4$.

- Find dA_t and dB_t .
- Write the corresponding explicit expressions for A_t and B_t :

$$\text{const} + \int_0^t \dots dW_u + \int_0^t \dots du$$

- Check whether X_t is a martingale.

3. Let $S_0 = 0, S_t = X_1 + X_2 + \dots + X_t$. The increments X_t are independent and identically distributed:

x	-1	0	1
$\mathbb{P}(X_t = x)$	0.2	0.2	0.6

- If possible find all constants a such that $M_t = S_t + at$ is a martingale.
- If possible find all constants b such that $R_t = b^{S_t}$ is a martingale.

4. Consider the process X_t

$$X_t = tW_t + \int_0^t u W_u^2 dW_u.$$

- Find $\mathbb{E}(X_t), \text{Var}(X_t)$.

(b) Find dX_t .

(c) Check whether X_t is a martingale.

5. A Hedgehog in the fog starts in $(0, 0)$ at $t = 0$ and moves randomly with equal probabilities in four directions (north, south, east, west) by one unit every minute.

Let X_t and Y_t be his coordinates after t minutes and $S_t = X_t + Y_t$.

(a) Find $\mathbb{E}(X_2 \mid S_2)$;

(b) Find $\text{Var}(X_2 \mid S_2)$.

Hint: $\text{Var}(Y \mid X) = \mathbb{E}(Y^2 \mid X) - (\mathbb{E}(Y \mid X))^2$.

6. Vampire Petr and Markov Chains.

Vampire Petr drinks blood of a new victim every day. Unfortunately 20% of the population are vaccinated against vampires. If more than one victim of the last three victims are vaccinated then Petr will be instantaneously cured and will return to the normal life.

For simplicity let's assume that the last three victims were not vaccinated.

(a) What is the probability that vampire Petr will be cured in the next three days?

(b) How many victims will be bitten by vampire Petr on average?

7. Vampire Boris and Martingales.

To survive vampire Boris needs to bite 70 talented students.

These 70 talented students have formed a secret group. They have written their emails on small pieces of paper and have randomly distributed these pieces among them. Each student has exactly one piece of paper with an email¹.

Initially vampire Boris knows contacts of just two persons from the group. Today he will contact them, drink their blood and get the emails they have. Then vampire Boris will contact new victims and so on.

(a) For $t \geq 1$ consider the process M_t , the proportion of non bitten students after the day t .

Is this process a martingale?

(b) Using martingale stopping theorem or otherwise find the probability that vampire Boris will bite all 70 students.

¹The group is so secret that it is possible that a student has his own email on his piece of paper

2 December exam

2.1 2025-2026b

Official chief mistress of King Louis XV Madame de Pompadour was born exactly 204 years earlier, on 29 December 1721. Her original name was Jeanne Antoinette Poisson.

1. Jeanne Antoinette Poisson receives letters from Diderot and d'Alembert. Each of them independently may be in a good mood or bad mood equiprobably. Every person sends letters to Jeanne Antoinette Poisson according to an independent Poisson process with intensity $\lambda_{\text{good}} = 2$ or $\lambda_{\text{bad}} = 1$ [letters per week] depending on his mood. Let (M_t) be the number of letters received by Madame de Pompadour in t weeks.

- (a) [2 + 3] Find $\mathbb{E}(M_t)$ and $\text{Var}(M_t)$.
- (b) [2] Is (M_t) a Poisson process? If so, find its intensity.
- (c) [3] Let T_1 be the time of the first letter. Find its cumulative distribution function $F(t)$.

2. We select a point (X, Y) randomly uniformly inside the square with vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, and $(0, 4)$.

- (a) [2 + 3] Find $\mathbb{E}(Y \mid Y - X)$ and $\text{Var}(Y \mid Y - X)$.
- (b) [2] Consider the σ -algebra $\mathcal{F} = \sigma(Y - X)$. Provide an example of two non-trivial events from $\mathcal{F}$ and two events not from $\mathcal{F}$.
- (c) [3] Find $\text{Cov}(X, Y \mid Y - X)$.

Hint: by definition $\text{Cov}(X, Y \mid \mathcal{F}) = \mathbb{E}(XY \mid \mathcal{F}) - \mathbb{E}(X \mid \mathcal{F}) \mathbb{E}(Y \mid \mathcal{F})$.

3. Let $S_0 = 0$. Each hour independently of previous history the process S_n increases by 1 with probability 0.25, decrease by 1 with probability 0.25 or does not change. Consider the first moment of time T when $|S_T| = 4$.

- (a) [3] If possible find α such that $Q_t = S_t^2 + \alpha t$ is a martingale.
- (b) [3] If possible find β such that $Q_t = \exp(S_t + \beta t)$ is a martingale.
- (c) [4] Find $\mathbb{E}(T)$.

4. Consider the standard Wiener process $(W_t)_{t \geq 0}$.

- (a) [4] Find $\text{Corr}(W_4 - W_2, W_3 - W_1)$.
- (b) [6] Find $\mathbb{E}(W_3 - W_1 \mid W_4 - W_2)$ and $\text{Var}(W_3 - W_1 \mid W_4 - W_2)$.

5. Consider three processes, $A_t = \int_0^t u \, dW_u$, $B_t = \int_0^t u W_u^4 \, dW_u$, $C_t = \int_0^t u W_u^2 \, du$.

- (a) [2 + 2] Find $\mathbb{E}(A_t)$, $\mathbb{E}(C_t)$.
- (b) [2 + 4] Find $\text{Var}(B_t)$, $\text{Cov}(A_t, C_t)$.

6. Be brave! Consider the stochastic process $X_t = g(t)(\exp(2026W_t) + \exp(-2026W_t))$ where $g(t)$ is some non-random function.

- (a) [6] Find dX_t and hence represent X_t as a sum of a constant, an Ito integral and a Riemann integral.
- (b) [4] If possible find any $g(t) \neq 0$ such that X_t is a martingale.

2.2 2025-2026a

Сегодня по народному календарю Евстратиев день. Нельзя в этот день сквернословить — не то ведьмы с неба упадут прямо на голову. Также нельзя веник на крыльце оставлять — не то ведьмы утащат.

1. The Legendary November Midterm Grader grades tests according to Poisson point process with variable intensity rate. The first 5 weeks the rate is equal to $\lambda_1 = 1$ test per week. Later the rate increases to $\lambda_2 = 100$ tests per week. Let X_t be the number of tests graded after t weeks.

- (a) [2] Sketch the curve $v(t) = \text{Var}(X_t)$.
- (b) [4] Using normal approximation find the probability that LNMG will check not less than 120 works at time $t = 6$.
- (c) [4] Now $t = 4.5$ weeks have already passed. Find the expected time for the next test checked.

Note: in point (b) you may use the standard normal cdf $F()$ in your answer.

2. The joint distribution of X and Y is given by the table:

	$Y = -1$	$Y = 0$	$Y = 1$
$X = -1$	1/8	1/16	1/8
$X = 0$	1/16	1/4	1/16
$X = 1$	1/8	1/16	1/8

Three students are building statistical models with different information sets:

- Roman Bokhyan: $\mathcal{F}_1 = \sigma(X, Y)$, the σ -algebra generated by X and Y .
- Mikhail N: $\mathcal{F}_2 = \sigma(X + Y, X - Y)$.
- Andrey L: $\mathcal{F}_3 = \sigma(X^2 + Y^2)$.

- (a) [3] For each pair of σ -algebras above determine whether one is contained in the other, they are equal, or neither. Justify your claims.
 - (b) [2] Andrey and Mikhail decide to share their information. Compute the number of elements in the smallest σ -algebra containing both $\mathcal{F}_2$ and $\mathcal{F}_3$.
 - (c) [5] Help Andrey and compute $\mathbb{E}(X \mid \mathcal{F}_3)$, $\text{Var}(X \mid \mathcal{F}_3)$.
3. The process S_n models the «accumulated wrath of the heavens». Let $S_0 = 0$. Each hour independently of previous history S_n increases by 1 with probability 1/3, decrease by 1 with probability 1/6 or does not change. The witches fall from the sky when S_n first reaches the level of 4. Let $T = \min\{n : S_n = 4\}$.
- (a) [4] For which $\alpha \neq 1$ is the process $M_n = \alpha^{S_n}$ a martingale?
 - (b) [2] Check whether $Q_n = S_n - n/6$ is a martingale.
 - (c) [4] Find $\mathbb{E}(T)$.

Hint: you may use Doob's theorem without checking technical details.

4. Consider three processes,

$$A_t = \int_0^t W_u^2 dW_u, \quad B_t = \int_0^t W_u^4 dW_u, \quad C_t = \int_0^t W_u^2 du.$$

- (a) [2 + 2] Find $\mathbb{E}(A_t)$, $\mathbb{E}(C_t)$.
- (b) [2 + 2 + 2] Find $\text{Var}(A_t)$, $\text{Cov}(A_t, B_t)$, $\text{Var}(C_t)$.

5. Be brave!

(a) [2 + 3] Find $\mathbb{E}(W_3^4)$, $\mathbb{E}(W_3^4 | W_2 = 10)$.

(b) [5] Find $\text{Cov}(W_1 W_2, W_5 W_6)$.

6. Consider the process

$$R_t = \exp \left(\int_0^t u^3 dW_u + t^2 \right).$$

(a) [6] Write R_t as a sum of a constant, an Ito's integral and a Riemann integral.

(b) [2] Is (R_t) a martingale?

(c) [2] Provide an ordinary (non-stochastic) differential equation for $h(t) = \mathbb{E}(R_t)$. You don't need to solve this equation.

Hint: you may write R_t as $R_t = \exp(Q_t)$ and apply Ito's lemma :)

2.3 2024-2025

In this text (W_t) denotes the standard Wiener process².

1. [10] We keep our promises!

(a) [3] What is the distribution of $W_7 + 2W_8$?

(b) [5] What is the conditional distribution of $(W_7 + 2W_8 | W_1 = 2)$?

(c) [2] Find the probability $\mathbb{P}(W_7 + 2W_8 > 1 | W_1 = 2)$ in terms of a standard normal cdf $F(\cdot)$.

2. [10] Consider processes $X_t = \int_0^t W_u^3 dW_u$ and $Y_t = \int_0^t W_u^4 du$.

(a) [3 + 3] Find $\mathbb{E}(X_t + Y_t)$ and $\text{Var}(X_t)$.

(b) [4] Find $\text{Cov}(X_t, W_t)$.

3. [10] Let $X_t = (W_t + g(t)) \exp(-W_t - t/2)$ with $X_0 = 0$.

(a) [4 + 2] Find dX_t and write X_t as a sum of two integrals.

(b) [4] Find at least one function $g(t)$ such that X_t is a martingale.

4. [10] It's the midterm exam. Five students are sitting in the last row. The student in the middle of the last row is the only one who brought a calculator.

Every minute his calculator moves one seat to the right (+1) or one seat to the left (-1) with equal probabilities.

Let's X_n be the coordinate of the calculator at time n with $X_0 = 0$.

(a) [4] Check whether $M_n = X_n^2 - n$ is a martingale.

(b) [6] Find the average time for the calculator to reach the end of the row (left or right).

5. [10] Consider a two-period binomial tree model with an initial share price $S_0 = 100$. The up and down share price multipliers are $u = 2$ and $d = 0.5$. Risk-free interest rate is $r = 10\%$ in the first period. The central bank will increase the interest rate in the second period exactly to $r = 20\%$.

The option is a European call with strike price $K = 300$ and maturity $T = 2$.

(a) [4] Find the risk neutral probabilities.

(b) [6] Find the arbitrage free price X_0 of this option.

6. [10] The assistant has not checked the home assignments in time. Hence teachers are receiving student complaints according to a Poisson process with daily rate $\lambda = 3$ messages per hour from 7 : 00 to 23 : 00 and nightly rate $\lambda = 1$ message per hour from 23 : 00 to 7 : 00.

Let X be the number of messages from 6 : 00 to 9 : 00 and Y — number of messages from 7 : 00 to 9 : 00.

(a) [2+2+2] Find $\mathbb{E}(X)$, $\text{Var}(X)$ and $\mathbb{P}(X = 2)$.

(b) [4] Find the conditional distribution of Y given that $X = 2$.

²Броуновское движение [ещё пока не запрещено на территории РФ]

2.4 2023-2024

1. [10 points] Let (X_t) be independent identically distributed random variables with $\mathbb{P}(X_i = -1) = 0.4$ and $\mathbb{P}(X_i = +1) = 0.6$. Consider the sum $S_t = X_1 + X_2 + \dots + X_t$.

- (a) [3] Is S_t a martingale?
 (b) [7] Find all constants c such that $M_t = c^{S_t}$ is a martingale.

2. [10 points] Let $a(t)$ be a deterministic function, $M_t = a(t) \cos(3W_t)$ and (W_t) is a Wiener process.

- (a) [4] Find dM_t .
 (b) [6] Find a non-zero function $a(t)$ such that (M_t) is a martingale.

3. [10 points] You have two correlated Wiener processes, (A_t) and (B_t) , with $\text{Corr}(A_t - A_s, B_t - B_s) = \rho$ for all $t > s$.

Split the time interval $[0; t]$ into n small segments of equal length. Let Δ_i^A be the increment of the Wiener process (A_t) on the i -th small segment, i.e. $\Delta_i^A = A(it/n) - A((i-1)t/n)$. Let Δ_i^B be the increment of the Wiener process (B_t) on the i -th small segment.

Consider the sum of cross-products, $S_n = \sum_{i=1}^n \Delta_i^A \Delta_i^B$.

- (a) [3] Find $\mathbb{E}(S_n)$.
 (b) [4] Does $\text{Var}(S_n)$ tend to 0 when $n \rightarrow \infty$?
 (c) [2] Find the mean square limit of S_n .
 (d) [1] How would you write this limit in a short hand notation with dA_t and dB_t ?
4. [10 points] The process (X_t) has $X_0 = 2024$, $dX_t = W_t^2 dW_t + W_t dt$, where (W_t) is a Wiener process.
- (a) [2] Is (X_t) a martingale?
 (b) [4] Find $d(X_t W_t)$.
 (c) [4] Find $\text{Cov}(X_t, W_t)$.

5. [10 points] Consider two-period binomial model with initial share price $S_0 = 600$. Up and down share price multipliers are $u = 1.2$, $d = 0.9$, risk-free interest rate is $r = 0.05$ per period.

The option pays you the maximal share price $X_2 = \max\{S_0, S_1, S_2\}$ at $t = 2$.

- (a) [4] Find the risk neutral probabilities.
 (b) [6] Find the current price X_0 of this option.
6. [10 points] Consider the framework of Black and Scholes model with riskless rate r , volatility σ and initial share price S_0 .

Find the current price X_0 of an option that pays you $X_T = 1$ at fixed time T if $S_T \geq 2S_0$.

Hint: you may use the standard normal cumulative distribution function in your answer.

2.5 2022-2023

Short rules: 120 minutes, you may use two A4 cheat-sheets, offline + one online group.

1. Consider $X_t = \int_0^t W_u^3 dW_u + \int_0^t (W_u^3 + 3W_u u) du - W_t^3 \cdot t$.

- (a) Find dX_t and the corresponding full form.
 (b) Is X_t a martingale?

2. Consider $X_t = \exp(-2W_t - 2t)$.

- (a) Find dX_t . Is X_t a martingale?
- (b) Find $\mathbb{E}(X_t)$ and $\text{Var}(X_t)$.
- (c) Find $\int_0^t X_u dW_u$.
3. As usual (W_t) is a Wiener process.
- (a) Find $\mathbb{E}(W_5 W_4 \mid W_4)$, $\text{Var}(W_5 W_4 \mid W_4)$.
- (b) Find covariance $\text{Cov}(W_4 W_5, W_5 W_6)$.
4. Let X_i be independent identically distributed with $\mathbb{P}(X_i = 1) = 0.9$, $\mathbb{P}(X_i = -1) = 0.1$.
Find all constants a and b such that $Y_t = a \exp(b \sum_{i=1}^t X_i)$ is a martingale.
5. Consider two-period binomial model with initial share price $S_0 = 600$, Up and down multipliers are $u = 1.2$, $d = 0.9$, risk-free interest rate is $r = 0.05$ per period.
Consider an option that pays you $X_2 = 100$ at $T = 2$ if $S_2 > S_1$ and nothing otherwise.
- (a) Find the risk neutral probabilities.
- (b) Find the current price X_0 of the asset.
- (c) How much shares should I have at $t = 1$ in the «up» state of the world to replicate the option?
6. Consider Black and Scholes model with riskless rate r , volatility σ and initial share price S_0 .
Find the current price X_0 of an option that pays you $X_2 = S_1^3$ at time $T = 2$.

2.6 2021-2022

Short rules: 120 minutes, online without proctoring, (W_t) is a standard Wiener process.

Date: 2021-12-25

1. (10 points) Consider an Ito's process $I_t = 2022 + W_t t^2 + \int_0^t W_u^3 dW_u + \int_0^t W_u^2 du$.
- (a) Find dI_t and check whether I_t is a martingale.
- (b) Check whether $J_t = I_t - \mathbb{E}(I_t)$ is a martingale.
2. (10 points) The random variables (Z_t) are independent identically distributed with moment generating function given by $M_Z(u) = 1/(1 - 5u)^3$.
We define X_t as $X_t = \exp(Z_1 + 2Z_2 + 3Z_3 + \dots + tZ_t)$ with $X_0 = 0$.
If possible find a martingale of the form $Y_t = h(t)X_t$ where $h(\cdot)$ is a non-random function.
3. (10 points) The process (Z_t) in discrete time is called *stationary* if it has constant expected value and constant covariances γ_k that do not depend on t .
- $$\begin{cases} \mathbb{E}(Z_t) = \mu; \\ \text{Cov}(Z_t, Z_t) = \gamma_0; \\ \text{Cov}(Z_t, Z_{t+1}) = \gamma_1; \\ \text{Cov}(Z_t, Z_{t+2}) = \gamma_2; \\ \dots \end{cases}$$
- (a) If possible provide an example of a martingale that is not stationary.
- (b) If possible provide an example of a stationary process that is not a martingale.
4. (10 points) Find $\mathbb{E}(W_1 W_2 W_3)$ and $\mathbb{E}(W_2 W_3 \mid W_1)$.

5. (10 points) Ded Moroz would like to receive $X_T = S_T^{-1}$ at time T if $S_T < 1$ and nothing otherwise. Assume the framework of Black and Scholes model, S_t is the share price, r is the risk free rate, σ is the volatility. How much Ded Moroz should pay now at $t = 0$?
6. (20 points) Martingales are everywhere :) Consider the process $Y_t = \exp(-uW_t)$.
- Find a multiplier $h(u, t)$ such that $M_t = h(u, t) \cdot Y_t$ is a martingale.
 - Find dY_t , $\mathbb{E}(Y_t)$ and $\text{Var}(Y_t)$.
 - Consider M_t that you have found as a function of u . Find the Taylor approximation of the function $M_t(u)$ up to u^4 .
 - Consider the coefficient before u^4 in the Taylor expansion of $M_t(u)$. Is it a martingale?
7. Bonus point. Guess your exam result (out of 70 possible points).

2.7 2020-2021

Today we celebrate Christmas Eve and 78 years of the Narkompros (People's Commissariat for Education) order governing the compulsory use of the letter «ё» in education process.

Date: 2020-12-24

- Ded Moroz would like to receive S_1^3 roubles at time $T = 2$, where S_t is the share price. Assume Black-Schöles model is valid, the risk-free rate is $r = 0.1$ and current share price is $S_0 = 100$. How much Ded Moroz should pay now at $t = 0$?
- Consider stationary $AR(2)$ model, $y_t = 2 + 0.3y_{t-1} - 0.02y_{t-2} + u_t$, where (u_t) is a white noise with $\text{Var}(u_t) = 4$. The last two observations are $y_{100} = 2$, $y_{99} = 1$.
 - Find 95% predictive interval for y_{102} .
 - Find the first two values of the autocorrelation function, ρ_1, ρ_2 .
 - Find the first two values of the partial autocorrelation function, ϕ_{11}, ϕ_{22} .

Hint: you need no more than 10 seconds to find both partial autocorrelations provided (b) is solved.

- The process y_t is described by a simple $GARCH(1, 1)$ model:

$$\begin{cases} y_t = \sigma_t \nu_t \\ \sigma_t^2 = 1 + 0.2y_{t-1}^2 + 0.3\sigma_{t-1}^2 \\ \nu_t \sim \mathcal{N}(0; 1) \end{cases}$$

The variables ν_t are independent of past variables $y_{t-k}, \nu_{t-k}, \sigma_{t-k}$ for all $k \geq 1$. The processes y_t, σ_t^2 are stationary.

Given $\sigma_{100} = 1$ and $\nu_{100} = 0.5$ find 95% predictive interval for y_{102} .

- Snegurochka studies a stochastic analog of the Fibonacci sequence

$$y_t = y_{t-1} + y_{t-2} + u_t,$$

where (u_t) is a white noise process.

- How many non-stationary solutions are there?
- What can you say about the number and the structure of the stationary solutions?
- Can Snegurochka find two starting constants $y_0 = c_0$ and $y_1 = c_1$ in such a way to make a solution stationary?

Be brave! There are two more exercises!

5. The semi-annual y_t is modelled by $ETS(AAA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 4) \\ s_t = s_{t-2} + 0.1u_t \\ b_t = b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + b_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + b_{t-1} + s_{t-2} + u_t \end{cases}$$

- (a) Given that $s_{100} = 2$, $s_{99} = -1.9$, $b_{100} = 0.5$, $\ell_{100} = 4$ find 95% predictive interval for y_{102} .
- (b) In this problem particular values of parameters are specified. And how many parameters are estimated in semi-annual $ETS(AAA)$ model before real forecasting?
6. The variables x_t take values 0 or 1 with equal probabilities. The variables u_t are normal $\mathcal{N}(0; 1)$. All variables are independent.

Consider the process $z_t = x_t(1 - x_{t-2})u_t$.

- (a) Find the covariance $\text{Cov}(z_t, z_s)$. Is the process z_t stationary?
- (b) Given that $z_{100} = 2.3$ find shortest predictive intervals for z_{101} and z_{102} with probability of coverage at least 95%.

Bënus: How many letters «ë» have you spotted?

3 April exam

3.1 2025-2026b

1. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 100) \\ b_t = 0.4b_{t-1} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.4b_{t-1} + 0.1u_t \\ y_t = \ell_{t-1} + 0.4b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 30$ and $b_{100} = 2$.

- (a) [5] Find the 95% predictive interval for y_{102} .
(b) [5] Find conditional probability $\mathbb{P}(y_{102} > 40 \mid \ell_{100}, b_{100})$ in terms of the standard normal distribution function.
2. The semi-annual y_t is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.5u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \end{cases}$$

with $\ell_0 = 20$, $s_0 = 5$ and $s_{-1} = -5$.

- (a) [4] Find best point forecasts for y_{100} and y_{101} .
(b) [6] Assume the first observation y_1 is missing in the dataset, but the second one is known, $y_2 = 30$. Find the best point forecast for y_1 .
3. Consider $MA(1)$ process $y_t = 10 + u_t + 5u_{t-1}$, where u_t is a white noise process with variance 20.
- (a) [5] Find $\mathbb{E}(y_t)$, $\text{Var}(y_t)$, $\text{Cov}(y_t, y_s)$.
(b) [1] Is the process (y_t) stationary?
(c) [1] Is the process (y_t) invertible with respect to (u_t) ?
(d) [3] If possible represent y_t in the form $y_t = \alpha + v_t + \beta v_{t-1}$ where α and β are some constant with $\beta \neq 5$ and (v_t) is a white noise.
4. Consider the equation $y_t = 1 + 1.2y_{t-1} - 0.5y_{t-2} + u_t - 0.8u_{t-1}$, where u_t is a white noise process with variance σ^2 .
- (a) [1] How many non-stationary solutions does this equation have?
(b) [2] How many stationary solutions of $MA(\infty)$ form with respect to (u_t) does this equation have?
(c) [2] Plot all roots of the AR lag polynomial on the complex plane and indicate the unit circle.
(d) [3] Rewrite all stationary solutions in $AR(\infty)$ form with respect to (u_t) , if possible.
(e) [2] Find $\mathbb{E}(y_t)$ for all stationary solutions.

5. Consider the stationary solution of the equation

$$y_t = 0.3y_{t-1} + 0.4y_{t-2} + u_t,$$

where (u_t) is a white noise process.

- (a) [3] How many stationary solutions does this equation have?
(b) [1] Check the invertibility condition for this equation.

- (c) [3] Find the values of ACF $\rho(1)$, $\rho(2)$, $\rho(3)$ for the stationary solution.
- (d) [3] Find the values of PACF $\phi(1)$, $\phi(2)$, $\phi(3)$ for the stationary solution.

6. For each of the following equations:

- Check whether a stationary solution exists.
- Check whether the process is invertible.
- Identify the process as $ARIMA(p, d, q)$ with the smallest possible d ($d = 0$ for stationary process).
- If the process is non-stationary, suggest a transformation to make it stationary.
- If the AR and MA polynomials have common roots, state them.

- (a) [2] $y_t = 0.5y_{t-1} + 0.5y_{t-2} + u_t - 0.5u_{t-1}$
- (b) [2] $y_t = 1.5y_{t-1} - 0.5y_{t-2} + u_t - 0.2u_{t-1}$
- (c) [3] $y_t = y_{t-1} - 0.25y_{t-2} + u_t - u_{t-1} + 0.25u_{t-2}$
- (d) [3] $(1 - L)^2 y_t = (1 - 0.8L)u_t$

3.2 2024-2025

1. [10] The process (u_t) is a white noise with $\text{Var}(u_t) = \sigma^2$. Consider the process

$$y_t = (2 + (-1)^t)u_1 + (3 + (-1)^t)u_2.$$

- (a) [4] Find $\mathbb{E}(y_t)$ and $\text{Var}(y_t)$.
- (b) [4] Find $\text{Cov}(y_t, y_s)$.
- (c) [2] Is the process (y_t) stationary?
2. [10] Consider the stationary solution of the equation $y_t = 2 + 0.9y_{t-1} + u_t - 0.5u_{t-1}$, where (u_t) is a white noise process with variance 60.
- (a) [4] If possible rewrite this solution as $AR(\infty)$ process.
- (b) [4] If possible rewrite this solution as $MA(\infty)$ process.
- (c) [2] Find $\text{Cov}(u_t, y_s)$ for this solution.
3. [10] Consider the equation $y_t = 7 + 0.4y_{t-1} - 0.13y_{t-2} + u_t + 2u_{t-1}$, where (u_t) is a white noise.
- (a) [1] How many non-stationary solutions does this equation have?
- (b) [4] How many stationary solutions of $MA(\infty)$ form with respect to (u_t) does this equation have?
- (c) [3] Can we rewrite the stationary solution in $AR(\infty)$ form with respect to (u_t) ?
- (d) [2] Find $\mathbb{E}(y_t)$ for the stationary solution.
4. [10] Let (y_t) be the solution of the equation $y_t = 2y_{t-1} - y_{t-2} + u_t$, where (u_t) are independent and normally distributed $\mathcal{N}(0; 9)$ and y_0 is a constant.
- (a) [5] Find 95% confidence for y_{101} given that $y_{100} = 3$ and $y_{99} = 4$.
- (b) [5] Find 95% confidence for y_{102} given that $y_{100} = 3$ and $y_{99} = 4$.
5. [10] For the stationary solution of the equation $y_t = 0.3y_{t-1} - 0.02y_{t-2} + u_t$, where (u_t) is a white noise process.
- (a) [5] Find the first three values of the autocorrelation function ρ_1, ρ_2, ρ_3 .
- (b) [5] Find all values of the partial autocorrelation function ϕ_{kk} .
6. [10] Let (y_t) be $MA(1)$ process.
- (a) [5] What are the possible values of $\rho_1 = \text{Corr}(y_t, y_{t-1})$?
- (b) [5] What are the possible values of the partial correlation $\phi_{22} = \text{pCorr}(y_t, y_{t-2}; y_{t-1})$?

3.3 2023-2024

Short rules: 120 minutes, one A4 cheat sheet allowed.

Date: 2024-03-05

1. Consider $MA(2)$ process given by

$$y_t = 5 + u_t + 2u_{t-1} + 4u_{t-2},$$

where (u_t) is a white noise with $\text{Var}(u_t) = \sigma^2$.

- (a) [1] Find the expected value $\mathbb{E}(y_t)$.
- (b) [7] Find the autocorrelation function $\rho_k = \text{Corr}(y_t, y_{t-k})$.
- (c) [2] Is the process (y_t) stationary?

2. Consider $MA(2)$ process given by

$$y_t = 5 + u_t + 2u_{t-1} + 4u_{t-2},$$

where u_t are normal independent random variables with $\text{Var}(u_t) = 4$.

You know that $u_{100} = 2$ and $u_{99} = -1$.

- (a) [5] Find the 95% predictive interval for y_{101} .
- (b) [5] Find the 95% predictive interval for $y_{1000001}$.

3. The stationary process (y_t) has autocorrelation function $\rho_k = 0.2^k$ and expected value 100.

- (a) [7] Find the first two values of the partial autocorrelation function, ϕ_{11} and ϕ_{22} .
- (b) [3] Provide a possible linear recurrence equation for this process. Your equation may include y_t , its lags and a white noise process (u_t) .

4. Consider the equation $y_t = 5 + 2.5y_{t-1} - y_{t-2} + u_t$, where (u_t) is a white noise process.

- (a) [3] Find the roots of the corresponding characteristic equation.
- (b) [4] Rewrite the process as $A(L)(y_t - \mu) = u_t$. You should explicitly write the lag polynomial $A(L)$ and the value of μ .
- (c) [1] How many non-stationary solutions does the equation have?
- (d) [1] How many stationary solutions does the equation have?
- (e) [1] How many stationary solutions of the $MA(\infty)$ form with respect to (u_t) does the equation have?

5. [10] The semi-annual (y_t) is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 4) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \end{cases}$$

Given that $s_{100} = 3$, $s_{99} = -2$, $\ell_{100} = 100$ find 95% predictive interval for y_{102} .

6. [10] The semi-annual (y_t) is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 4) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \\ \ell_0 = 100, s_0 = -3, s_{-1} = 3 \end{cases}$$

Check whether the process (y_t) is stationary.

3.4 2022-2023

Short rules: 90 minutes, one A4 cheat sheet allowed.

Date: 2023-03-25

1. Consider $ETS(AAdN)$ model

$$\begin{cases} u_t \sim \mathcal{N}(0; 20) \\ b_t = 0.9b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + 0.9b_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + 0.9b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 20$ and $b_{100} = 1$.

- Find 95% prediction interval for y_{102} .
- Approximately find the best point forecast for y_{10000} .

2. Consider the difference equation:

$$y_t = 0.7y_{t-1} - 0.12y_{t-2} + u_t,$$

where (u_t) is a white noise.

- How many stationary and non-stationary solutions does the difference equation have?

Consider stationary $AR(2)$ process that satisfies the difference equation.

- Find first two values of autocorrelation function.
- Find α_1 and α_2 in $MA(\infty)$ representation

$$y_t = u_t + \alpha_1 u_{t-1} + \alpha_2 u_{t-2} + \alpha_3 u_{t-3} + \dots$$

3. The strictly stationary white noise (u_t) follows $ARCH(1)$ model $\sigma_t^2 = 3 + 0.5u_{t-1}^2$ where $u_t = \sigma_t \nu_t$ and $\nu_t \sim \mathcal{N}(0; 1)$.

- Find 95% prediction interval for u_{101} given that $u_{100} = -1$.
- Find $\mathbb{E}(u_t)$, $\text{Var}(u_t)$.
- Find $\text{Corr}(u_t, u_{t-1})$, $\text{Corr}(u_t^2, u_{t-1}^2)$.

4. The weight of a fish Y_i is a discrete random variables with distribution and observed frequencies given in the table

Weight [kg]	1	2	4
Probability	$0.2 + a$	$0.3 - a$	0.5
Observed frequency	N_1	N_2	N_4

Fish weights Y_i are independent.

- Find the maximum likelihood estimator of the parameter a .
 - Find the method of moments estimator of the parameter a .
5. You observe time between taxi arrivals on a stop, $Y_1, Y_2, \dots, Y_n$. Assume that Y_i are independent and exponentially distributed with $\mathbb{E}(Y_i) = \theta$, that means the density of each Y_i is $f(y) = \exp(-y/\theta)/\theta$ for $y \geq 0$. Consider the following estimator of expected value

$$\hat{\theta} = n \cdot \min\{Y_1, Y_2, \dots, Y_n\}$$

- Find the probability density function of $\hat{\theta}$.
- Is $\hat{\theta}$ unbiased?
- Is $\hat{\theta}$ consistent?

3.5 2021-2022

Short rules: 120 minutes, one A4 cheat sheet allowed.

Date: 2022-04-04

1. Consider $ETS(AAN)$ model,
$$\begin{cases} y_t = \ell_{t-1} + b_{t-1} + u_t \\ \ell_t = \ell_{t-1} + b_{t-1} + \alpha u_t \\ b_t = b_{t-1} + \beta u_t \\ u_t \sim \mathcal{N}(0; \sigma^2). \end{cases}$$

Let $\ell_{100} = 50$, $b_{100} = 2$, $\alpha = 0.4$, $\beta = 0.5$, $\sigma^2 = 16$.

Calculate one step and two steps ahead 95% predictive intervals.

2. Consider the process $y_t = 4 + u_t + u_{t-1} + 2u_{t-2}$, where (u_t) is a white noise with variance 16.
- Is this process stationary? Explain.
 - Find the autocorrelation function of this process. Explain the meaning of ρ_2 .
 - Consider the process $d_t = \Delta y_t$. Is it $ARIMA(p, d, q)$? If yes, then find p , d and q .
3. Consider the stationary $AR(2)$ process $y_t = 5 - 0.9y_{t-1} - 0.2y_{t-2} + u_t$, where (u_t) is a white noise.
- Find the first value of autocorrelation function ρ_1 .
 - Find the partial autocorrelation function of this process. Explain the meaning of ϕ_{22} .
 - What is the relationship between values of autocorrelation function ρ_{100} , ρ_{99} and ρ_{98} .

Hint: values ϕ_{22} , ϕ_{33} etc may be calculated almost effortlessly :)

4. Consider iid sample from bivariate normal distribution, $\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \theta \\ 2\theta \end{pmatrix}; \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}\right)$.

Calculate Fischer information for the following cases:

- You observe X_1 only.
- You observe $X_1, \dots, X_n$.
- You observe $X_1, \dots, X_n, Y_1, \dots, Y_n$.

Hint: the multivariate normal density is $f(u) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp(-\frac{1}{2}(u - \mu)^T \Sigma^{-1}(u - \mu))$.

5. Random variables $X_1, \dots, X_n$ are independent with density $f(x) = \begin{cases} -\ln(a) \cdot a^x, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$
- Estimate a using maximum likelihood.
 - Check whether the estimator is unbiased and consistent.
 - Check whether the corresponding Cramer-Rao lower bound is attained.
6. Consider the $ARCH(1)$ model, $u_t = \sigma_t \nu_t$, where ν_t are iid $\mathcal{N}(0; 1)$ and $\sigma_t^2 = 1 + 0.3u_{t-1}^2$.
- Find 95% predictive interval for u_{101} if $u_{100} = -2$.
 - Find the autocorrelation function of $r_t = u_t^2$.

3.6 2020-2021

Date: 2021-04-13, Rock 'N' Roll day

Estimation questions

1. To go to the mountain top I use a gondola lift in the morning. I go back from the top using the same gondola lift in the evening. Cabins are numbered from 1 to a .

I have noticed that the absolute difference of cabin numbers of my two trips was 10.

- (a) Estimate a using maximum likelihood.
- (b) Estimate a using method of moments.

2. Random variables $X_1, X_2, \dots, X_n$ are independent identically distributed with density

$$f(x_i | \lambda, a) = \frac{\lambda}{2} \exp(-\lambda|x_i - a|).$$

Observed values for $n = 3$ are $-3, 1, 11$.

- (a) Estimate λ using method of moments for fixed $a = 1$.
- (b) Estimate λ and a using maximum likelihood.

3. Random variables $X_1, \dots, X_n$ are independent and normally distributed $\mathcal{N}(1, 1/b)$.

- (a) Estimate b using maximum likelihood.
- (b) Does the estimator achieve the Cramer-Rao lower bound?
- (c) Is the estimator consistent?
- (d) Is the estimator unbiased?

4. Random variables $X_1, X_2, \dots, X_n$ are independent identically distributed with density

$$f(x_i | \lambda) = \frac{\lambda}{2} \exp(-\lambda|x_i|).$$

For $n = 100$ I have 40 negative values with sum equal to -300 and 60 positive values with sum equal to 500.

- (a) Test the hypothesis $\lambda = 1$ using LR approach at significance level $\alpha = 0.01$.
- (b) Test the hypothesis $\lambda = 1$ using LM approach at significance level $\alpha = 0.01$.

Distribution questions

5. I have three problems in the home assignment. Time spent on each problem is modelled by independent exponentially distributed random variables with rate λ : X_1, X_2, X_3 .

- (a) Find the moment generating function of X_i and hence the moment generating function of $S = X_1 + X_2 + X_3$.
- (b) Find $\mathbb{E}(S^3)$.
- (c) Find the joint density of $R = X_1/(X_1 + X_2 + X_3)$ and S .

6. I have 100 numbers written on small sheets of paper: $x_1, x_2, \dots, x_{100}$. The sum of these numbers is 1.

Find the possible values of the sum

$$\frac{x_1}{\sqrt{1-x_1}} + \frac{x_2}{\sqrt{1-x_2}} + \dots + \frac{x_{100}}{\sqrt{1-x_{100}}}.$$

Hint: consider a randomly selected number X and apply the Jensen's inequality.

4 Final exam

4.1 2025-2026b

1. Let $x_1, x_2, \dots, x_n$ be a random sample from a Gamma distribution with density

$$f(x | \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0,$$

where $\alpha > 0$ is the shape parameter and $\beta > 0$ is the scale parameter.

It is known that $\mathbb{E}(x_i) = \alpha\beta$ and $\mathbb{E}(x_i^2) = \alpha(\alpha + 1)\beta^2$.

- (a) [4] Find the method of moments estimators for α and β .
 - (b) [2] For the observed sample (2, 1, 3, 2, 4), compute the numerical values of $\hat{\alpha}_{MM}$ and $\hat{\beta}_{MM}$.
 - (c) [3] Are the estimators $\hat{\alpha}_{MM}$ and $\hat{\beta}_{MM}$ consistent?
 - (d) [1] Assume now that the shape parameter α is known. Find the method of moments estimator for β and check whether it is unbiased.
2. Let $x_1, x_2, \dots, x_n$ be a random sample from the Pareto distribution with probability density function

$$f(x | \alpha) = \frac{\alpha \cdot 2^\alpha}{x^{\alpha+1}}, \quad x \geq 2,$$

where $\alpha > 0$ is the shape parameter.

- (a) [4] Derive the maximum likelihood estimator $\hat{\alpha}_{MLE}$ for the parameter α .
 - (b) [2] Verify that the obtained estimator corresponds to a global maximum of the likelihood function.
 - (c) [1] Find the sufficient statistic for α .
 - (d) [3] Check whether $\hat{\alpha}_{MLE}$ is unbiased for α .
3. Let $y_1 = \theta + \varepsilon_1$ and $y_2 = 2\theta + \varepsilon_2$, where $\varepsilon_1 \sim \mathcal{N}(0, 1)$ and $\varepsilon_2 \sim \mathcal{N}(0, 4)$ are independent random variables.

We are interested in estimating unknown $q = \theta^2$ based on the observations y_1, y_2 . Consider quadratic estimators of the form

$$\hat{q} = c_1 y_1^2 + c_2 y_2^2 + d,$$

where c_1, c_2, d are some coefficients.

- (a) [1] Find the distribution of y_1 and y_2
 - (b) [3] Find the condition on c_1, c_2, d such that the estimator $\hat{q}$ is unbiased for q .
 - (c) [3] Assuming the unbiasedness conditions, find the $\text{MSE}(\hat{q})$. Hint: $\mathbb{E}(y_i^4) = i^4(\theta^4 + 6\theta^2 + 3)$.
 - (d) [3] Let $\theta = 2$. Find the values of c_1, c_2 and d that minimize the $\text{MSE}(\hat{q})$ subject to the unbiasedness condition.
4. Consider the following joint distribution of the vector (X_i, Y_i) ,

$$\mathbb{P}(X_i = x, Y_i = y) = p(1-p)^{y-x}q(1-q)^x,$$

where $x \in \{0, 1, 2, \dots\}$, $y \in \{0, 1, 2, \dots\}$ and $x \leq y$. Observations $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independent.

- (a) [5] Construct a likelihood ratio test for the hypothesis $H_0: p = 1/2$ against alternative $p \neq 1/2$ if you know that $q = 1/2$.
- (b) [5] Construct a likelihood ratio test for the hypothesis $H_0: p = q$ against alternative $p \neq q$.

Reminder: clearly specify the distribution of LR -statistic under H_0 and describe the decision rule.

5. Consider the following joint distribution of the vector (X_i, Y_i) ,

$$\mathbb{P}(X_i = x, Y_i = y) = p(1-p)^{y-x}q(1-q)^x,$$

where $x \in \{0, 1, 2, \dots\}$, $y \in \{0, 1, 2, \dots\}$ and $x \leq y$. Observations $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independent.

- (a) [5] Find theoretical Fisher information for unknown p if you know that $q = 1/2$.
- (b) [5] Find theoretical Fisher information for unknown p and q .

6. Consider the following joint distribution of the vector (X_i, Y_i) ,

$$\mathbb{P}(X_i = x, Y_i = y) = p(1-p)^{y-x}q(1-q)^x,$$

where $x \in \{0, 1, 2, \dots\}$, $y \in \{0, 1, 2, \dots\}$ and $x \leq y$. Observations $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independent.

- (a) [5] Find any minimal sufficient statistic for unknown p if you know that $q = 1/2$.
- (b) [5] Find any minimal sufficient statistic for unknown p and q .

4.2 2025-2026a

Date: 2026-04-24. Twenty years ago youtube published the first video «me at the zoo».

1. Suppose that salaries $x_1, x_2, \dots, x_n$ (measured in units of the minimum wage) are independent and follow a simple Pareto distribution with probability density function

$$f(x | \alpha) = \frac{\alpha}{x^{\alpha+1}}, \quad x \geq 1,$$

where $\alpha > 1$ is an unknown shape parameter.

- (a) [4] Find the method of moments estimator $\hat{\alpha}_{MM}$.
- (b) [5] Let $p \in (0, 1]$ be the proportion of the population with the highest salaries. Show that the share of total income received by this top p fraction of the population is

$$D(p) = p^{\frac{\alpha-1}{\alpha}}.$$

- (c) [1] The empirical Pareto principle (the «80/20 rule») states that approximately 20% of the population receives 80% of the total income. Using the formula find $\hat{\alpha}_{rule}$ that satisfies

$$D(0.2) = 0.8.$$

2. Let $x_1, x_2, \dots, x_n$ be a random sample from a binomial distribution $\text{Bin}(N, p)$ with probability mass function

$$P(x_i = x) = C_N^x p^x (1-p)^{N-x}, \quad x = 0, 1, 2, \dots, N.$$

- (a) [4] Find the method of moments estimators for N and p .

Consider known $N = 4$ and the observed sample $(4, 1, 2, 2, 1)$.

- (b) [3] Find the MLE estimate of p .
- (c) [2] Prove that MLE estimator of p is unbiased and consistent.
- (d) [1] Find a minimal sufficient statistics for p .

3. The depth of corrosion μ [mm] on a metal plate is measured using four different unbiased independent methods (ultrasonic, electrical resistance, optical, and X-ray). Due to the different principles and equipment, the measurement variances are known to be

$$\sigma_1^2 = 1.6, \quad \sigma_2^2 = 2, \quad \sigma_3^2 = 2.5, \quad \sigma_4^2 = 3.$$

The measured values [mm] of obtained by the four methods are

$$x_1 = 19, \quad x_2 = 18, \quad x_3 = 20, \quad x_4 = 21.$$

- (a) [3] Give at least one example of an unbiased estimator $\tilde{\mu}$ of μ .
 - (b) [4] Find the linear unbiased estimator $\hat{\mu}$ of μ that has the smallest possible variance.
 - (c) [1] Compute the numerical value of $\hat{\mu}$ using the given data.
 - (d) [2] Compute the standard error of $\hat{\mu}$.
4. Random variables $X_1, \dots, X_n$ are independent with Poisson distribution $\mathbb{P}(X_i = k) = \exp(-\lambda)\lambda^k/k!$ for non-negative integers k . Random variables $Y_1, \dots, Y_n$ are independent with $\mathbb{P}(Y_i = 1) = p$ and $\mathbb{P}(Y_i = 0) = 1 - p$. I do not observe X_i nor Y_i , I observe only their product $W_i = X_i Y_i$. Among $n = 1000$ observations of W_i : 300 observations are exactly equal to 0 and $\sum W_i = 7000$.

Test the null hypothesis $p = 1$ against alternative $p < 1$ using likelihood ratio test with significance level 0.05.

FYI: right 0.05 critical chi-squared values are $\chi_{1,\text{crit}}^2 = 3.84$, $\chi_{1,\text{crit}}^2 = 5.99$, $\chi_{1,\text{crit}}^2 = 7.81$.

5. Random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed with

$$\mathbb{P}(X_i = 0) = \exp(-a), \quad \mathbb{P}(X_i = k) = \mathbb{P}(X_i = -k) = \exp(-a)a^k/(2 \cdot k!) \text{ for } k \in \mathbb{N}.$$

Let $Y_i = |X_i|$.

- (a) [6] Calculate Fisher information for the sample $X_1, \dots, X_n$ and for the sample $Y_1, \dots, Y_n$. Which sample contains more information about unknown a ?
 - (b) [4] Does maximum likelihood estimator based on $X_1, \dots, X_n$ attain the Cramer-Rao boundary?
6. Random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed with

$$\mathbb{P}(X_i = 0) = \exp(-a), \quad \mathbb{P}(X_i = k) = \mathbb{P}(X_i = -k) = \exp(-a)a^k/(2 \cdot k!) \text{ for } k \in \mathbb{N}.$$

- (a) [4] Find minimal sufficient statistic T for the unknown parameter a .

Let $g(X_1, X_2, \dots, X_n)$ be some function that does not depend on a directly.

- (b) [3] Is it possible that $\mathbb{E}(T \mid g(X_1, X_2, \dots, X_n))$ depends on a ?
- (c) [3] Is it possible that $\mathbb{E}(g(X_1, X_2, \dots, X_n) \mid T)$ depends on a ?

4.3 2024-2025

1. Unobserved random variables (y_i) are independent and uniform on $[-2a, a]$ with unknown $a > 0$.
 - (a) [4] Find $\mathbb{E}(y_i^2)$ and $\mathbb{E}(|y_i|)$.
 - (b) [3] Find the method of moments estimator of a if you know the value of $\sum y_i^2$.
 - (c) [3] Find the method of moments estimator of a if you know the value of $\sum |y_i|$.
2. Dragon Erik receives gifts from two kingdoms: $x_1, x_2, \dots, x_{10}$ from Ex-Kingdom and $y_1, y_2, \dots, y_{20}$ from Why-Kingdom. Variables (x_i) are exponentially distributed with rate λ_x and variables (y_i) — with rate λ_y . All variables are independent. Erik has two hypothesis, $H_0: \lambda_x = \lambda_y$ and $H_1: \lambda_x \neq \lambda_y$. In the observed sample $\sum x_i = 300$, $\sum y_i = 500$.

- (a) [4] Find the unrestricted maximum likelihood estimates of λ_x and λ_y .
- (b) [3] Find the maximum likelihood estimate of the rate under H_0 .
- (c) [3] Test H_0 against H_1 at 0.05 significance level using likelihood ratio test.

Note: $\mathbb{P}(\chi_1^2 \geq 3.84) = 0.05$, $\mathbb{P}(\chi_2^2 \geq 5.99) = 0.05$, $\mathbb{P}(\chi_3^2 \geq 7.81) = 0.05$.

3. The variables (X_i) are independent and have exponential distribution with unknown rate λ .

Yuki-the-Hedgehog would like to estimate the unknown parameter $a = 1/\lambda^2$. He uses the estimator

$$\hat{a} = \frac{X_1^2 + X_2^2 + \cdots + X_n^2}{2n + 1}$$

- (a) [5] Is the estimator unbiased?
- (b) [5] Is the estimator consistent?

4. Princess Lika Pankova would like to test the hypothesis H_0 that y_1 and y_2 are independent and uniform on $[0, 1]$ against the alternative that they have the joint density $f(y_1, y_2) = 2 - y_1 - y_2$ for $y_1, y_2 \in [0, 1]$.

Lika accepts only the best!

- (a) [7] Construct the test with 0.1 probability of the first type error and lowest probability of the second type error.
- (b) [3] What is the minimal achieved second type error probability?
5. Arina loves all kinds of raisin-studded Easter bread, but her grandmother's is her absolute favorite. Grandmother's Easter bread sizes (y_i) are independent identically distributed with density $f(y) = ab^a/y^{a+1}$ for $y \geq b$ and zero otherwise.
- (a) [2] Draw a typical density for some values of parameters a and b .
- (b) [4] Find a sufficient statistic for the unknown a given that $b = 1$ kilo.
- (c) [4] Find a sufficient statistic for the unknown a and b .

6. The semi-annual (y_t) is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 9) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \end{cases}$$

Let $s_{100} = 2$, $s_{99} = -3$, $\ell_{100} = 200$.

- (a) [6] Find 95% predictive interval for y_{102} .
- (b) [4] Write this model in the form $A(L)y_t = B(L)u_t$, where $A(L)$ and $B(L)$ are lag polynomials.

4.4 2023-2024

Short rules: 120 minutes, you may use one A4 cheat-sheet, offline

Date: 2024-04-27

1. The variables $X_1, \dots, X_n$ are independent identically distributed with density

$$f(x) = \begin{cases} \lambda \exp(-\lambda(x - \theta)), & \text{if } x \geq \theta \\ 0, & \text{otherwise.} \end{cases}$$

- (a) [5] Find the method of moments estimator of λ for known value $\theta = 1$ using the first moment.
- (b) [5] Find the method of moments estimator of λ for unknown value θ using the first two moments.
2. [10] The variables $X_1, \dots, X_n$ are independent and normally distributed $\mathcal{N}(a, 2a)$.
Find the maximum likelihood estimator of a .
Hint: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-(x - \mu)^2/2\sigma^2)$.
3. The variables $X_1, \dots, X_n$ are independent and uniformly distributed $U[0; a]$ with $a > 1$. We do not observe X_i directly but we know whether each X_i is larger than 1. Hence we observe the indicators $Y_i = I(X_i > 1)$.
Consider the estimator $\hat{a} = 1/(1 - \bar{Y})$.
- (a) [5] Is $\hat{a}$ consistent?
- (b) [5] Is $\hat{a}$ unbiased for $n = 2$?

4. The variables $X_1, \dots, X_n$ are independent and have Poisson distribution with intensity rate λ . In other words the probability mass function is given by $\mathbb{P}(X_i = k) = \exp(-\lambda)\lambda^k/k!$.

- (a) [5] Find theoretical Fisher information for λ contained in the sample.
- (b) [2] Derive the maximum likelihood estimator for λ .
- (c) [3] Does the maximum likelihood estimator attain the Cramer-Rao lower bound for variance?
5. [10] The variables $X_1, \dots, X_n$ are independent and gamma distributed with density

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} \exp(-\lambda x) / \Gamma(\alpha), & \text{if } x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) [5] Find a sufficient statistic for α if we know that $\lambda = 1$.
- (b) [5] Find a two dimensional sufficient statistic for unknown α and λ .
6. We have two independent random samples $X_1, X_2, \dots, X_{n_x}$ and $Y_1, Y_2, \dots, Y_{n_y}$. The random variables X_i follow Poisson distribution with intensity rate λ_x , random variables Y_i follow Poisson distribution with intensity rate λ_y .
- We would like to test $H_0: \lambda_x = \lambda_y$ against $H_1: \lambda_x \neq \lambda_y$.

- (a) [3] Find the maximal value of log-likelihood under H_0 .
- (b) [3] Find the maximal value of log-likelihood under unrestricted model.
- (c) [2] Construct the likelihood ratio test.
- (d) [2] Do you reject H_0 if $n_x = 100$, $n_y = 200$, $\sum x_i = 500$, $\sum y_i = 900$ at significance level 5%?

Hint: chi-squared critical values for $\alpha = 0.05$ are $\chi_{df=1}^2 = 3.84$, $\chi_{df=2}^2 = 5.99$.

4.5 2022-2023

Short rules: 120 minutes, you may use one A4 cheat-sheet, offline + online.

Notes: W_t denotes the standard Wiener process, you may use standard normal cumulative distribution function in your answers.

Date: Balalayka day, 2023-06-23.

1. The weight of a fish Y_i is a discrete random variables with distribution and observed frequencies given in the table

Weight [kg]	1	2	a
Probability	$0.2 + 0.1a$	$0.3 - 0.1a$	0.5
Observed frequency	N_1	N_2	N_a

Fish weights Y_i are independent, $a > 10$ is unknown.

- (a) Find the method of moments estimator of the parameter a .
- (b) Find the maximum likelihood estimator of the parameter a .
2. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 20) \\ b_t = 0.9b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + 0.9b_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + 0.9b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 20$ and $b_{100} = 2$.

- (a) Find conditional probability $\mathbb{P}(y_{102} > 30 \mid \ell_{100}, b_{100})$.
- (b) Approximately find the best point forecast for y_{10000} .

3. Stochastic process X_t is defined by $X_t = 7 + u_t + 0.3u_{t-1}$, where (u_t) is a white noise with variance σ^2 .
 - (a) Is (X_t) stationary?
 - (b) Find the autocorrelation function of (X_t) .
 - (c) Find $\mathbb{E}(X_{t+2} \mid X_t, X_{t-1}, \dots)$.
4. Consider the process $X_t = \int_0^t W_u^2 dW_u + \int_0^t (W_u^2 + 2W_u u) du - W_t^2 \cdot t$.
 - (a) Find dX_t and the corresponding full form.
 - (b) Is X_t a martingale?
 - (c) Find $\mathbb{E}(X_t)$.
5. Consider the Black and Scholes model with riskless rate r , volatility σ and initial share price S_0 .
Find the current price X_0 of an option that pays you one dollar at time $T = 2$ only if $S_2 > \exp(3r)S_0$.
6. A hedgehog moves at random on the vertices A, B, C and D of a regular tetrahedron (тетраэдр). She start at the vertex A and every minute changes her position to one of the adjacent vertices with probability $1/3$ independently of past moves.
 - (a) Write down the transition matrix of this Markov chain.
 - (b) What is the expected time of the first return to the starting vertex A ?

4.6 2021-2022

Short rules: 120 minutes, offline, one A4 cheat sheet allowed.

Date: 2022-06-25

1. Consider $ETS(ANN)$ model,
$$\begin{cases} y_t = \ell_{t-1} + u_t \\ \ell_t = \ell_{t-1} + \alpha u_t \\ u_t \sim \mathcal{N}(0; \sigma^2). \end{cases}$$
 Let $\ell_{99} = 50$, $\alpha = 1/2$, $\sigma^2 = 16$, $y_{98} = 48$, $y_{99} = 52$, $y_{100} = 55$. Calculate 95% predictive interval for y_{101} .
2. Young investor Winnie-the-Crypto compares two trading strategies: buying bitcoins from good bees and from bad bees. Let d_t be the price difference at day t (bad minus good). Winnie-the-Crypto would like to test $H_0: \mathbb{E}(d_t) = 0$ against $H_a: \mathbb{E}(d_t) \neq 0$ at 5% significance level.
Winnie assumed that (d_t) can be approximated by a $MA(1)$ process and estimated the parameters using $T = 400$ observations, $\hat{d}_t = 2 + u_t + 0.7u_{t-1}$ with $\hat{\sigma}_u^2 = 4$.
 - (a) Estimate $\mathbb{E}(d_t)$, $\text{Var}(d_t)$ and $\text{Cov}(d_t, d_{t-1})$.
 - (b) Estimate $\mathbb{E}(\bar{d})$, $\text{Var}(\bar{d})$ and help Winnie by considering $Z = \frac{\bar{d} - 0}{\text{se}(\bar{d})}$.
3. The variables $X_1, \dots, X_n$ are independent and uniformly distributed on $[0; 2a]$ for some positive a .
 - (a) Find any sufficient statistic for a .
 - (b) How the answer will change if $X_i \sim U[-a; 2a]$?
4. Consider an estimator $\hat{a}$ with $\mathbb{E}(\hat{a}) = 0.5a + 3$. For the given sample size the Fisher information is $I_F(a) = 400/a^2$.
 - (a) What is the theoretical minimal variance of $\hat{a}$?
 - (b) Assume that $\hat{a}$ attains the minimal variance boundary and is asymptotically normal. Given that $\hat{a} = 2022$ provide 95% CI for a .

5. You observe $X_1, \dots, X_{400}$ and $Y_1, \dots, Y_{400}$, $\bar{X} = 5$, $\bar{Y} = 6$. All variables are independent.

Consider the null hypothesis that all random variables are exponentially distributed with common parameter λ against alternative that parameter is λ_X for every X_i and λ_Y for every Y_j .

- Estimate common λ using maximum likelihood for the restricted model.
 - Estimate both λ_X and λ_Y using maximum likelihood in the unrestricted model.
 - Use LR-test to test the null hypothesis at 5% significance level.
6. The ultimate goal of this exercise is to prove the good upper bound for tail probability of a normal distribution: if $X \sim \mathcal{N}(0; \sigma^2)$ then $\mathbb{P}(X > c) \leq \exp(-c^2/2\sigma^2)$.

Here are the guiding hints (you free to use not use them):

- State the MGF of X . You may derive it or simply write it if you remember.
 - Consider $Y = \exp(uX)$. Using Markov inequality provide the upper bound for $\mathbb{P}(Y > \exp(uc))$.
 - Prove that $\mathbb{P}(X > c) \leq MGF_X(u) \exp(-uc)$ for any u .
 - Find the value of u that makes the upper bound as tight as possible.
7. (bonus) Draw good bees and bad bees selling crypto. Any funny statistics/math joke is also ok!

4.7 2020-2021

Today: $+31^\circ$, World Refrigeration Day :)

You have 100 minutes. You can use A4 cheat sheet and calculator. Be brave!

Date: 2021-06-26

- I throw a fair die until the sequence 626 appears. Let N be the number of throws.
 - What is the expected value $\mathbb{E}(N)$?
 - Write down the system of linear equations for the moment generating function of N . You don't need to solve it!
- Consider the following stationary process

$$y_t = 1 + 0.5y_{t-2} + u_t + u_{t-1},$$

where random variables u_t are independent $\mathcal{N}(0; 4)$.

- Find the 95% predictive interval for y_{101} given that $y_{100} = 2$, $y_{99} = 3$, $y_{98} = 1$, $u_{99} = -1$.
 - Find the point forecast for y_{101} given that $y_{100} = 2$.
- I have an unfair coin with probability of heads equal to $h \in (0; 1)$.
 - Let N be the number of tails before the first head. Find the MGF of N .
 - Let S be the number of tails before k heads (not necessary consecutive). Find the MGF of S .
 - What is the limit of $MGF_S(t)$ when $k \rightarrow \infty$ and $k \times h \rightarrow 0.5$? What is the name of the corresponding distribution?
 - Consider the stochastic process $X_t = f(t) \cos(2021W_t)$.
 - Find dX_t .
 - Find any $f(t) \neq 0$ such that X_t is a martingale.
 - Using $f(t)$ from the previous point find $\mathbb{E}(\cos(2021W_t))$.

5 October exam solutions

5.1 2025-2026

5.2 2024-2025

5.3 2023-2024

1. (a) $\mathbb{P}(S_3 = B) = \mathbb{P}(A \rightarrow B \rightarrow A \rightarrow B) + \mathbb{P}(A \rightarrow C \rightarrow A \rightarrow B) + \mathbb{P}(A \rightarrow B \rightarrow C \rightarrow B)$
 (b)

$$\begin{pmatrix} 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{pmatrix}$$

(c)

$$\begin{cases} a = 0.5b + 0.5c \\ b = 0.7a + 0.5c \\ c = 0.3a + 0.5b \\ a + b + c = 1 \end{cases}$$

The solution is $a = 15/45$, $b = 17/45$, $c = 13/45$.

2. (a) Consider N as known fixed value, $\mathbb{E}(S \mid N) = N \mathbb{E}(X_1) = N \cdot 0.5$. First, let's find moment generating function for X_i :

$$M_X(u) = \int_0^1 \exp(xu) \cdot 1 \, dx = \frac{\exp(u) - 1}{u};$$

Hence $M_{S|N}(u) = (M_X(u))^N$ as S is the sum of N independent variables.

- (b) Random variable N is discrete, $M_S(u) = \mathbb{P}(N = 1)(M_X(u))^1 + \mathbb{P}(N = 2)(M_X(u))^2 + \dots = \frac{0.7M_X(u)}{1 - 0.3M_X(u)}$.

- (c) Moment generating function is used to calculate moments, $M_S''(0) - (M_S'(0))^2 = \text{Var}(S)$.

3. Start with X_0 : $\mathbb{E}(X_0) = 1$, $\text{Var}(X_0) = 4/12 = 1/3$.

- (a) Expected value is constant, $\mathbb{E}(X_n) = 0.5 + 0.5 \mathbb{E}(X_{n-1})$, hence $\mathbb{E}(X_n) = 1$. Variance goes to zero, $\text{Var}(X_n) = 0.25 \text{Var}(X_{n-1})$.

- (b) $\text{plim } X_n = 1$

4. Let's measure time in minutes.

- (a) $\mathbb{P}(X_5 = 0) = \exp(-5\lambda) = 0.05$, so $\lambda = \ln(0.05)/-5 = \ln(20)/5$.

- (b) $\mathbb{E}(X_{180}) = 180\lambda$, $\text{Var}(X_{180}) = 180\lambda$

- (c) $\mathbb{P}(X_{180} = 5) = \exp(-180\lambda)(180\lambda)^5/5!$

5. (a) $Y = IX_1 + (1 - I)X_2$

- (b) Consider I as known or fixed variable, $\mathbb{E}(Y \mid I) = I \mathbb{E}(X_1) + (1 - I) \mathbb{E}(X_2)$. Note that $I^2 = I$ and $(1 - I)^2 = 1 - I$, hence $\text{Var}(Y \mid I) = I \text{Var}(X_1) + (1 - I) \text{Var}(X_2)$.

- (c) $\mathbb{E}(Y) = p\mu_1 + (1 - p)\mu_2$ and $\text{Var}(Y) = p(1 - p)(\mu_1 - \mu_2)^2 + p\sigma_1^2 + (1 - p)\sigma_2^2$, where $p = 0.3$, $\mu_1 = \sigma_1^2 = 1$, $\mu_2 = \sigma_2^2 = 2$.

6. (a) Sigma-algebra: $\sigma(X) = \{\emptyset, \Omega, \{X = -2\}, \{X = 0\}, \{X = 2\}, \{X \neq -2\}, \{X \neq 0\}, \{X \neq 2\}\}$. Other descriptions are possible, for example, one may replace $\{X = -2\}$ by $\{X < 0\}$.

- (b) Random variable XY takes 3 distinct values, hence $\text{card } \sigma(X \cdot Y) = 2^3 = 8$.

5.4 2022-2023

1.

$$\begin{aligned} \text{plim} \frac{\sum (X_i - \bar{X})^3}{n + 2022} &= \text{plim} \frac{\sum X_i^3 - 3\bar{X} \sum X_i^2 + 3\bar{X}^2 \sum X_i - \sum \bar{X}^3}{n + 2022} = \\ &= \mathbb{E}(X_1^3) - 3\mathbb{E}(X_1^2) + 3\mathbb{E}(X_1) - 1 = 0; \end{aligned}$$

Note that $\mathbb{E}(X_1^2) = 4/3$, $\mathbb{E}(X_1^3) = 2$.

2. $\mathbb{P}(X_3 = 1) = 0.3 \cdot 0.7 \cdot 0.3 + 0.7 \cdot 1 \cdot 0.3 = 0.21 \cdot 1.3$ Let's denote $\tau_j = \min\{t \mid X_t = 0, X_0 = j\}$, $\mu_j = \mathbb{E}(\tau_j)$.

$$\begin{cases} \mu_0 = 0 \\ \mu_1 = 1 + 0.7\mu_2 \\ \mu_2 = 1 + 0.3\mu_1 + 0.7\mu_3 \\ \mu_3 = \mu_2 + 1 \end{cases}$$

We get $\mu_2 = 200/9$.

3. (a) $\sigma(X_1 \cdot X_2) = \{\emptyset, \Omega, \{X_1 X_2 = 1\}, \{X_1 X_2 = -1\}\}$;
 (b) Many answers are ok, for example $\sigma(X_1 = X_3)$.
 (c) Note that $\sigma(X_1, X_1 + X_2, X_1 + X_2 + X_3) = \sigma(X_1, X_2, X_3)$, the number of events in sigma-algebra is $\text{card } \sigma(X_1, X_1 + X_2, X_1 + X_2 + X_3) = 2^8 = 256$.
4. $\text{Cov}(N_3, N_{10}) = \text{Cov}(N_3, N_3 + (N_{10} - N_3)) = \text{Var}(N_3) = 3\lambda$.
5. $\mathbb{E}(X_3 \mid X_1, X_2) = \mathbb{E}(X_3) = 0$, $\mathbb{E}(X_3 \mid X_1 + X_3) = (X_1 + X_3)/2$, $\text{Var}(X_3 \mid X_1, X_3) = 0$, $\text{Var}(X_3 \mid X_1 + X_3) = 1 - (X_1 + X_3)^2/4$.
 Посчитаем ожидание и получим $Y_n = X_1 + X_2 + \dots + X_n$, the process (Y_n) is a martingale.
6. $\text{Var}(Y_t - Y_s) = \text{Var}(tW_{2t} - sW_{2s}) = 2t^3 + 2s^3 - 4ts^2$. We get $\mathbb{E}(M_{t+u} \mid \mathcal{F}_t) = W_t^3 + 3W_t u + \alpha(t+u)W_t$.
 From $\mathbb{E}(M_{t+u} \mid \mathcal{F}_t) = W_t^3 + \alpha t W_t$ it follows that $\alpha = -3$.

5.5 2021-2022

1.

5.6 2021-2022 retake

1.

5.7 2020-2021

1.

6 December exam solutions

6.1 2025-2026b

- 1.
2. (a) $\mathbb{E}(Y | Y - X) = (4 + Y - X)/2$ and $\text{Var}(Y | Y - X) = (4 - |Y - X|)^2/12$.
 (b) From $\mathcal{F}$: $A_1 = \{X > Y\}$, $A_2 = \{X > Y + 0.5\}$. Not from $\mathcal{F}$: $B_1 = \{X > 2\}$, $B_2 = \{Y < 3\}$.
 (c) Let's just add a known value $Y - X$ to the first term, $\text{Cov}(X, Y | Y - X) = \text{Cov}(X + (Y - X), Y | Y - X) = \text{Var}(Y | Y - X)$.
3. (a) $\alpha = -0.5$;
 (b) $\beta = \ln 4 - \ln(e + e^{-1} + 2)$.
 (c) $\mathbb{E}(T) = 2 \cdot 4^2 = 32$.
4. (a) $\text{Corr}(W_4 - W_2, W_3 - W_1) = 1/\sqrt{2 \cdot 2} = 0.5$.
 (b) Let's find best linear prediction of $W_3 - W_1$ given $W_4 - W_2$, $\beta = \text{Cov}(W_4 - W_2, W_3 - W_1)/\text{Var}(W_4 - W_2) = 0.5$, hence $W_3 - W_1 = 0.5(W_4 - W_2) + R$.
 It follows that

$$\mathbb{E}(W_3 - W_1 | W_4 - W_2) = 0.5(W_4 - W_2)$$
 and that

$$\text{Var}(W_3 - W_1 | W_4 - W_2) = \text{Var}(R) = 2 - 0.25 \cdot 2 = 1.5.$$
5. (a) $\mathbb{E}(A_t) = 0$, $\mathbb{E}(C_t) = t^3/3$.
 (b) $\text{Var}(B_t) = 15t^7$, $\text{Cov}(A_t, C_t) = \dots$
6. From ordinary differential equation $g'(t) = -2026^2 g(t)$ we get for example, $g(t) = \exp(-2026^2/2t)$.

6.2 2025-2026a

- 1.
- 2.
- 3.
4. (a) $\mathbb{E}(A_t) = 0$, $\mathbb{E}(C_t) = t^2/2$.
 (b) $\text{Var}(A_t) = t^3$, $\text{Cov}(A_t, B_t) = 15t^4/4$, $\text{Var}(C_t) = 7t^4/12 - (t^2/2)^2$.

$$\mathbb{E}(C_t^2) = \int_0^t \int_0^t \mathbb{E}(W_u^2 W_s^2) du ds = 2 \int_{s=0}^t \int_{u=0}^s 2u^2 + us du ds = 2 \cdot 7/24t^4.$$
5. (a) $\mathbb{E}(W_3^4) = 3^2 \cdot 3 = 27$, $W_3 = W_2 + I$ where $I \sim \mathcal{N}(0, 1)$, hence $W_3^4 = (W_2 + I)^4$ and $\mathbb{E}(W_3^4 | W_2 = 10) = 10^4 + 6 \cdot 10^2 + 3$.
 (b) Using Isserlis theorem, $\text{Cov}(W_1 W_2, W_5 W_6) = \mathbb{E}(W_1 W_2 W_5 W_6) - \mathbb{E}(W_1 W_2) \mathbb{E}(W_5 W_6) = (5+2+2) - 5 = 4$.
6. Let $Q_t = \int_0^t u^3 dW_u + \int_0^t 2u du$, so $R_t = \exp(Q_t)$. By Ito's lemma:

$$dR_t = \exp(Q_t) dQ_t + 0.5 \exp(Q_t) (dQ_t)^2 = \exp(Q_t) (t^3 dW_t + 2t dt) + 0.5 \exp(Q_t) t^6 dt.$$

The differential equation is $h'(t) = h(t)(2t + t^6/2)$.

6.3 2024-2025**6.4 2023-2024**

1. (a) [3] S_t is not a martingale, $\mathbb{E}(S_{t+1} | \mathcal{F}_t) = S_t + 0.6 - 0.4 \neq S_t$.
 (b) [7] Two solutions, one is trivial $c = 1$, the other...

2. [10 points]

- (a) [4] $dM_t = -3a(t) \sin(3W_t) dW_t + a'(t) \cos(3W_t) dt - 0.5 \cdot 9a(t) \cos(3W_t) dt$
 (b) [6] If (M_t) is a martingale then $a'(t) - 4.5a(t) = 0$, hence $a(t) = c \cdot \exp(4.5t)$.

3. [10 points]

- (a) [3] $\mathbb{E}(S_n) = \rho t$.
 (b) [4] $\text{Var}(S_n) \rightarrow 0$
 (c) [2] $S_n \rightarrow \rho t$ in mean squared sense.
 (d) [1] $dA_t dB_t = \rho dt$

4. (a) [2] (X_t) is not a martingale, as we have $W_t dt$ term;
 (b) [4]

$$d(X_t W_t) = W_t dX_t + X_t dW_t + dX_t dW_t = (W_t^3 dW_t + W_t^2 dt) + X_t dW_t + W_t^2 dt = (W_t^3 + X_t) dW_t + 2W_t^2 dt$$

- (c) [4] $\text{Cov}(X_t, W_t) = \mathbb{E}(X_t W_t) - 0$,

$$X_t W_t = 0 + \int_0^t (W_u^3 + X_u) dW_u + \int_0^t 2W_u^2 du$$

$$\mathbb{E}(X_t W_t) = \int_0^t \mathbb{E}(2W_u^2) du = \int_0^t 2u du = t^2.$$

5. [10 points]

6. [10 points]

$$X_0 = \exp(-rT) \mathbb{E}^*(X_T) = \exp(-rT) \mathbb{P}^*(S_T \geq 2S_0) = \dots$$

6.5 2022-2023

1. (a) $dX_t = (W_t^3 - 3W_t^2 \cdot t) dW_t$ (4 points), 1 point for comment how you get the answer (definition and Ito's lemma), 2 points for full form

$$X_t = X_0 + \int_0^t (W_u^3 - 3W_u^2 \cdot u) dW_u$$

- (b) A process is a martingale as in short form $A_t dt = 0$ (3 points)

2. (a) $dX_t = -2X_t dW_t$ (2 points), this process is a martingale (1 point)

- (b) $\mathbb{E}(X) = 1$ (2 points), $\text{Var}(X) = \exp(4t) - 1$ (2 points)

(c)

$$\int_0^t X_u dW_u = \frac{1 - X_t}{2}$$

(3 points)

3. (a) 2 points for $\mathbb{E}(W_5 W_4 | W_4) = W_4^2$, 3 points for $\text{Var}(W_5 W_4 | W_4) = W_4^2$

- (b) i. 1-2 points for clever ideas
 ii. 3 points for solution with serious mistakes

iii. 4 points for solutions with arithmetic errors

iv. 5 points for $\text{Cov}(W_5 W_4, W_5 W_6) = 40$

4. 1-3 points depending on the cleverness of ideas.

5 points if one got correct martingale:

$$\mathbb{E}(Y_{t+1} \mid Y_t) = Y_t \mathbb{E}(e^{bX_{t+1}})$$

10 points if one solved equation correctly:

$$\mathbb{E}(e^{bX_{t+1}}) = 1 \rightarrow b = 0 \text{ or } b = \ln(1/9)$$

Minus 1 point if one forgot trivial solution $a = 0$ and any b .

5. (a) $p_u^* = p_d^* = 1/2$ (3 points)

(b) $X_1^u = X_1^d = (0.5 \cdot 100 + 0.5 \cdot 0)/1.05$, hence $X_0 = 50/1.05^2 \approx 45.35$ (3 points)

(c) $\alpha = X_2^{uu} - X_2^{ud} / (S_2^{uu} - S_2^{ud}) = 100/216 \approx 0.46$ (4 points)

6. You get 2 points almost for nothing:

$$X_0 = \exp(-2r) \mathbb{E}_*(X_2)$$

Correct formula for X_2 in terms of W_1^* gives your 4 points:

$$X_2 = S_1^3 = S_0^3 \exp(3r) \exp(3\sigma W_1^* - 9\sigma^2/2).$$

Calculations of expected value (4 points more):

$$X_0 = S_0^3 \exp(r) \exp(3\sigma^2).$$

6.6 2021-2022

1.

6.7 2020-2021

1.

7 April exam solutions

7.1 2025-2026b

1. $y_{102} = 31.12 + 0.14u_{101} + u_{102}$, $PI = [31.12 - 1.96\sqrt{101.96}; 31.12 + 1.96\sqrt{101.96}]$, $\mathbb{P}(A) = 1 - F(0.88)$.

2. $\hat{y}_{100} = 25$, $\hat{y}_{101} = 15$; $y_1 = 15 + u_1$, $y_2 = 25 + 0.5u_1 + u_2$,

$$\mathbb{E}(y_1 | y_2) = 15 + \text{Cov}(y_1, y_2) / \text{Var}(y_2)(y_2 - 25).$$

3. $\mathbb{E}(y_t) = 10$,

$$\gamma_k = \begin{cases} 520, & \text{if } k = 0, \\ 100, & \text{if } k = 1, \\ 0, & \text{otherwise.} \end{cases}$$

(y_t) is stationary but not invertible with respect to (u_t) .

$$y_t = 10 + v_t + 0.2v_{t-1}$$

7.2 2024-2025

7.3 2023-2024

1. (a) $\mathbb{E}(y_t) = 5$

(b) $\rho_3 = \rho_4 = \dots = 0$

(c) The process is stationary.

2.

3. (a) $\phi_{11} = \rho_1 = 0.2$, $\phi_{22} = 0$

(b) Possible equation is $y_t = 0.2y_{t-1} + u_t$. Another possibility is $y_t = 5y_{t-1} + u_t$. In the second case the stationary solution will be forward-looking and not $MA(\infty)$ with respect to (u_t) .

4. (a) $\lambda_1 = 2$, $\lambda_2 = 0.5$, here the roots of the lag polynomial are exactly the same.

(b) $(1 - 2L)(1 - 0.5L)(y_t + 10) = u_t$

(c) The equation has infinitely many non-stationary solutions.

(d) The equation has unique stationary solution.

(e) The equation has no stationary solutions that are $MA(\infty)$ with respect to (u_t) .

5.

6. The process is not stationary as $\mathbb{E}(y_1) = 3$ and $\mathbb{E}(y_2) = -3$.

7.4 2022-2023

1. (a) [6 points]

$$y_{102} = \ell_{100} + (0.9 + 0.9^2)b_{100} + (0.3 + 0.18)u_{101} + u_{102}$$

$$(y_{102} | y_1, \dots, y_{100}) \sim \mathcal{N}(21.71, 24.608)$$

The interval

$$[21.71 - 1.96 \cdot 4.96; 21.71 + 1.96 \cdot 4.96]$$

(b) [4 points]

$$\lim_{h \rightarrow \infty} \mathbb{E}(y_{100+h} | y_1, \dots, y_{100}) = \ell_{100} + (0.9 + 0.9^2 + \dots)b_{100} = 20 + 9 \cdot 1$$

2. (a) [2 points] $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, one stationary solution, infinitely many non-stationary solutions.

- (b) [6 points]: [2 points] for the system + [2 points] for ρ_1 + [2 points] for ρ_2 .

$$\begin{cases} \gamma_1 = 0.7\gamma_0 - 0.12\gamma_1 \\ \gamma_2 = 0.7\gamma_1 - 0.12\gamma_0. \end{cases}$$

$$\rho_1 = 70/112 = 0.625, \quad \rho_2 = 49/112 - 0.12 = 0.3175$$

- (c) [2 points]

$$\alpha_1 = 0.7, \quad \alpha_2 = 0.37$$

3. (a) [4 points]

$$\sigma_{101}^2 = 3 + 0.5(-1)^2 = 3.5$$

$$(u_{101} \mid \sigma_{101}) \sim \mathcal{N}(0; \sigma_{101}^2)$$

$$[-1.96\sqrt{3.5}; +1.96\sqrt{3.5}]$$

- (b) [3 points] [1 point] for $\mathbb{E}(u_t)$ and [2 points] for $\text{Var}(u_t)$ The process (u_t) is a white noise, hence

$$\mathbb{E}(u_t) = 0.$$

$$\sigma_u^2 = 3 + 0.5 \cdot \sigma_u^2$$

- (c) [3 points]: [1 point] for $\text{Corr}(u_t, u_{t-1})$ and [2 points] for $\text{Corr}(u_t^2, u_{t-1}^2)$ The process (u_t) is a white noise, hence

$$\text{Corr}(u_t, u_{t-1}) = 0.$$

$$u_t^2 = 3 + 0.5u_{t-1}^2 + (u_t^2 - \sigma_t^2)$$

We notice that $r_t = u_t^2 - \sigma_t^2$ is a white noise, hence u_t^2 is an $AR(1)$ process. Hence, $\text{Corr}(u_t^2, u_{t-1}^2) = 0.5$.

4. (a) [5 points]

$$L = \text{const}(0.2 + a)^{N_1}(0.3 - a)^{N_2}0.5^{N_3}$$

$$\ell = \text{const} + N_1 \ln(0.2 + a) + N_2 \ln(0.3 - a) + N_3 \ln 0.5$$

$$\frac{\partial \ell}{\partial a} = \frac{N_1}{0.2 + a} - \frac{N_2}{0.3 - a}$$

$$\hat{a}_{ML} = \frac{0.3N_1 - 0.2N_2}{N_1 + N_2}$$

We see that $\partial \ell / \partial a$ decreases as a increases, so $\hat{a}_{ML}$ is indeed the point of maximum.

- (b) [5 points]

$$\mathbb{E}(Y_i) = (0.2 + a) + 2(0.3 - a) + 4 \cdot 0.5 = 2.8 - a$$

$$\bar{Y} = \frac{N_1 + 2N_2 + 4N_4}{N_1 + N_2 + N_4}$$

$$\hat{a}_{MM} = 2.8 - \frac{N_1 + 2N_2 + 4N_4}{N_1 + N_2 + N_4}$$

5. (a) [6 points]

$$\mathbb{P}(\hat{\theta} > y) = \mathbb{P}(Y_1 > y/n)^n = (\exp(-y/n\theta))^n = \exp(-y/\theta)$$

Hence $\hat{\theta}$ has exponential distribution with rate $1/\theta$ and probability density function

$$f(t) = \begin{cases} \exp(-t/\theta)/\theta, & \text{if } t \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (b) [2 points]

The estimator is unbiased as

$$\mathbb{E}(\hat{\theta}) = 1/(1/\theta) = \theta.$$

- (c) [2 points]

The estimator is non consistent as its distribution does not depend on n .

7.5 2021-2022

- 1.
2. (a) Yes, the process is stationary, that is $MA(2)$ process.
(b) $\rho_3 = \rho_4 = \dots = 0$
(c) $d_t = u_t + u_{t-1} + 2u_{t-2} - u_{t-1} - u_{t-2} - 2u_{t-3}$, hence $d_t \sim ARIMA(0, 0, 3)$.
3. (a)
(b) $\phi_{11} = \rho_1$, $\phi_{22} = -0.2$, $\phi_{33} = \phi_{44} = \dots = 0$. The partial correlation ϕ_{22} measures how will y_t on average react to the unit change of y_{t-2} given fixed y_{t-1} .
(c) $\rho_{100} = -0.9\rho_{99} - 0.2\rho_{98}$

7.6 2020-2021

- 1.

8 Final exam solutions

8.1 2025-2026b

- 1.
- 2.
- 3.
- 4.
- 5.
6. $T_1 = \sum(Y_i - X_i)$, $T_2 = (\sum Y_i, \sum X_i)$ or any other equivalent.

8.2 2025-2026a

1. Threshold income t for rich people can be found from equation $\mathbb{P}(X > t) = p$. Average income is given by $\mathbb{E}(X)$. Average income of a rich guy is given by $\mathbb{E}(X | X > t) = \mathbb{E}(XI(X > t))/p$. Hence, proportion of income is given by

$$D(p) = \frac{\mathbb{E}(X | X > t) \cdot p}{\mathbb{E}(X)} = \frac{\mathbb{E}(XI(X > t))}{\mathbb{E}(X)}.$$

- 2.
- 3.
4. Let N_0 be the number of zeros in the sample. Likelihood is given by

$$L = (1 - p + p \exp(-\lambda))^{N_0} \prod_{W_i > 0} p \exp(-\lambda) \frac{\lambda^{W_i}}{W_i!}.$$

Recall that $\lambda^0 = 1$ and $0! = 1$ by gentleman's agreement. Hence, $\sum W_i = \sum_{W_i > 0} W_i$ and $\sum \ln W_i = \sum_{W_i > 0} \ln W_i$.

$$\ell(p, \lambda) = N_0(1 - p + p \exp(-\lambda)) + (n - N_0) \ln p - (n - N_0)\lambda + \ln \sum W_i - \sum \ln W_i!.$$

Restricted estimates are $\hat{p}_R = 1$, $\hat{\lambda}_R = \sum W_i / n = 7$. Unrestricted estimators can't be obtained in a closed form, but can be calculated approximately with sufficient precision. While solving the first order conditions we will get the equation

$$\exp(-\hat{\lambda}_{ur}) = 1 - \hat{\lambda}_{ur} / 10.$$

Exponential function decays very quickly, so $\hat{\lambda}_{ur} \approx 10$. For $\hat{\lambda} = 10$ the left hand side will be $\exp(-10)$ and the right hand side will be exactly 0. It follows that $\hat{p}_{ur} \approx 0.7$.

We use χ^2 with $2 - 1 = 1$ degree of freedom. The null hypothesis is rejected.

5. Let's compare likelihood for X_i and $Y_i = |X_i|$. As $Y_i \sim \text{Pois}(\lambda)$,

$$L_Y(a) = \prod \exp(-a) a^{Y_i} / Y_i!$$

Be careful with X_i . Let N_0 be the number of zeros in the sample. Also note that if $X_i < 0$ then you should take $(-X_i)!$.

$$L_X(a) = \exp(-a)^{N_0} \prod_{X_i \neq 0} \exp(-a) a^{|X_i|} / (2|X_i|!).$$

Rearrange a little bit!

$$L_X(a) = 2^{N_0 - n} \prod \exp(-a) a^{|X_i|} / |X_i|!$$

So, $\ln L_X(a)$ and $\ln L_Y(a)$ differs by $(N_0 - n) \ln 2$. All derivatives with respect to a are equal. Samples of X_i and Y_i have the same Fisher information, same sufficient statistics, same maximum likelihood estimators.

$$\hat{a}_{ml} = \sum Y_i / n = \sum |X_i| / n.$$

As $\text{Corr}(\hat{a}_{ml}, \ell') = 1$ the Cramer-Rao boundary is attained.

6. Sufficient statistic is $\sum |X_i| = \sum Y_i$. If you take $g(X_1, \dots, X_n)$ identically equal to 67 then $\mathbb{E}(T | g) = \mathbb{E}(T) = n \cdot a$ and we see that $\mathbb{E}(T | g)$ can depend on a . It is also possible that $\mathbb{E}(T | g)$ does not depend on a , take for example $g = T$.

$$\mathbb{E}(g | T) = \int_{\mathbb{R}^n} g(x_1, \dots, x_n) f(x_1, \dots, x_n | t) dx_1 dx_2 \dots dx_n.$$

Inside the integral nothing depends on a so $\mathbb{E}(g | T)$ can't depend on a .

8.3 2024-2025

8.4 2023-2024

1. Let's observe that we may decompose X_i as a sum $X_i = Y_i + \theta$, where $Y_i \sim \text{Exp}(\lambda)$.
Hence, $\mathbb{E}(X_i) = 1/\lambda + \theta$, $\text{Var}(X_i) = \text{Var}(Y_i) = 1/\lambda^2$ and $\mathbb{E}(X_i^2) = 1/\lambda^2 + (1/\lambda + \theta)^2$.
There is an alternative solution with direct integration:

$$\mathbb{E}(X_i) = \int_{\theta}^{+\infty} x f(x) dx, \quad \mathbb{E}(X_i^2) = \int_{\theta}^{+\infty} x^2 f(x) dx.$$

- (a) Solving $1/\hat{\lambda} + 1 = \bar{X}$ we obtain $\hat{\lambda} = 1/(\bar{X} - 1)$.
(b) Solving for $\hat{\lambda}$ and $\hat{\theta}$ the system

$$\begin{cases} 1/\hat{\lambda} + \hat{\theta} = \bar{X} \\ 1/\hat{\lambda}^2 + (1/\hat{\lambda} + \hat{\theta})^2 = M_2 \text{ with } M_2 = \sum X_i^2/n \end{cases}$$

we obtain

$$\hat{\lambda} = \frac{1}{\sqrt{M_2 - \bar{X}^2}}, \quad \hat{\theta} = \bar{X} - \sqrt{M_2 - \bar{X}^2}$$

2. The log-likelihood function is equal to

$$\ell(a) = \sum_{i=1}^n ((-0.5) \ln(4\pi) - 0.5 \ln a - (x_i - a)^2/4a).$$

The equation $\ell'(a) = 0$ may be simplified to

$$n\hat{a}^2 + 2n\hat{a} - \sum X_i^2 = 0$$

Hence,

$$\hat{a} = \frac{-2n \pm \sqrt{4n^2 + 4n \sum X_i^2}}{2n}$$

We choose the root $\hat{a} > 0$ as $\text{Var}(X_i) = 2a > 0$.

$$\hat{a} = \sqrt{1 + \sum X_i^2/n} - 1$$

Just for fun. In the case $X_i \sim \mathcal{N}(a, ka)$ the equation would be

$$n\hat{a}^2 + kn\hat{a} - \sum X_i^2 = 0$$

And

$$\hat{a} = \frac{-nk + \sqrt{k^2n^2 + 4n \sum X_i^2}}{2n}.$$

3. $\mathbb{E}(Y_i) = \mathbb{P}(X_i > 1) = (a - 1)/a = p$.

- (a) The estimator is consistent as

$$\text{plim } \hat{a} = \frac{1}{1 - \text{plim } \bar{Y}} = \frac{1}{1 - \frac{a-1}{a}} = a$$

- (b) For $n = 2$ we have the positive probability p^2 that $\bar{Y} = 1$. Hence with positive probability $\hat{a}$ is not defined. The value $\mathbb{E}(\hat{a})$ does not exist for $n = 2$.

4. (a) The log-likelihood function is equal to

$$\ell(\lambda) = \sum_{i=1}^n (-\lambda + X_i \ln \lambda - \ln(X_i!))$$

The score function is

$$\text{score}(\lambda) = \ell'(\lambda) = \sum_{i=1}^n (-1 + X_i/\lambda).$$

And

$$\ell''(\lambda) = \sum_{i=1}^n (-X_i/\lambda^2).$$

Fisher information is

$$I_F = -\mathbb{E}(\ell''(\lambda)) = \sum \mathbb{E}(X_i)/\lambda^2 = n\lambda/\lambda^2 = n/\lambda.$$

- (b) Solving $\ell' = 0$ we obtain

$$\hat{\lambda} = \bar{X}$$

- (c) Rewrite $\ell'(\lambda)$ using $\hat{\lambda}$. Be careful! Do not confound λ and $\hat{\lambda}$.

$$\text{score}(\lambda) = \ell'(\lambda) = -n + n\hat{\lambda}/\lambda.$$

Hence the score function is linear function of $\hat{\lambda}$, $\text{Corr}(\text{score}(\lambda), \hat{\lambda}) = 1$ and the Cramer-Rao bound is attained.

One may also find $\mathbb{E} \hat{\lambda} = \lambda$, $\text{Var}(\hat{\lambda}) = \lambda/n$ and explicitly check that the general bound

$$\text{Var}(\hat{\lambda}) \geq 1/I_F$$

is attained as equality in our case

$$\lambda/n = 1/(n/\lambda).$$

5. We do not need the formula for $\Gamma(\alpha)$ here.

- (a) For known $\lambda = 1$ the likelihood is

$$L = \left(\prod X_i \right)^{\alpha-1} \frac{1}{\Gamma(\alpha)} \cdot \exp(-\sum X_i).$$

If we optimize this function for α the optimal $\hat{\alpha}$ will depend only on $\prod X_i$. Hence $\prod X_i$ is a sufficient statistic for α . There are many other sufficient statistics, $\sum \ln X_i$ is another example.

- (b) Now the likelihood is

$$L = \left(\prod X_i \right)^{\alpha-1} \frac{1}{\Gamma(\alpha)} \lambda^\alpha \exp(-\lambda \sum X_i).$$

If we optimize this function for α and λ the optimal point will depend only on $\prod X_i$ and $\sum X_i$. Hence $(\prod X_i, \sum X_i)$ is a two dimensional sufficient statistic for (α, λ) .

6. (a) Under H_0 we have $X_i \sim \text{Pois}(\lambda)$, $Y_i \sim \text{Pois}(\lambda)$.

$$\ell(\lambda) = \sum_{i=1}^{n_x} (-\lambda + X_i \ln \lambda - \ln(X_i!)) + \sum_{i=1}^{n_y} (-\lambda + Y_i \ln \lambda - \ln(Y_i!))$$

The score function is

$$\text{score}(\lambda) = \ell'(\lambda) = \sum_{i=1}^{n_x} (-1 + X_i/\lambda) + \sum_{i=1}^{n_y} (-1 + Y_i/\lambda).$$

The estimator is $\hat{\lambda} = (\sum X_i + \sum Y_i)/(n_x + n_y)$.

$$\max \ell_R = -\hat{\lambda}(n_x + n_y) + \left(\sum X_i + \sum Y_i \right) \ln \hat{\lambda} - \sum \ln X_i! - \sum \ln Y_i!$$

- (b) In unrestricted model we have two independent estimators,

$$\hat{\lambda}_x = \bar{X}, \quad \hat{\lambda}_y = \bar{Y}$$

$$\max \ell_{UR} = -\hat{\lambda}_x n_x + \sum X_i \ln \hat{\lambda}_x + \sum Y_i \ln \hat{\lambda}_y - \sum \ln X_i! - \sum \ln Y_i!$$

- (c)

$$LR = 2(\max \ell_{UR} - \max \ell_R) = 2 \sum X_i (\ln \hat{\lambda}_x - \ln \hat{\lambda}) + 2 \sum Y_i (\ln \hat{\lambda}_y - \ln \hat{\lambda})$$

- (d) Unrestricted model has two parameters, restricted model has one parameter, hence we use chi-squared distribution with $2 - 1 = 1$ degree of freedom, $LR_{\text{crit}} = 3.84$. We calculate estimates, $\hat{\lambda}_x = 5$, $\hat{\lambda}_y = 4.5$, $\hat{\lambda} = 14/3$.

$$LR = 1000(\ln 5 - \ln(14/3)) + 1800(\ln 4.5 - \ln(14/3)) \approx 3.5$$

We do not reject H_0 .

8.5 2022-2023

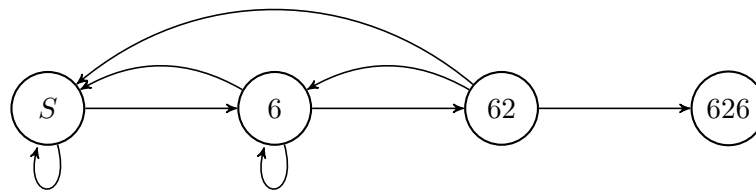
- 1.

8.6 2021-2022

- 1.

8.7 2020-2021

1. Let's draw the chain



The system of equations for expected values:

$$\begin{cases} x_s = 1 + \frac{1}{6}x_6 + \frac{5}{6}x_s \\ x_6 = 1 + \frac{1}{6}x_6 + \frac{1}{6}x_{62} + \frac{4}{6}x_s \\ x_{62} = 1 + \frac{1}{6} \cdot 0 + \frac{1}{6}x_6 + \frac{4}{6}x_s \end{cases}$$

The system of equations for moment generating functions:

$$\begin{cases} m_s(t) = \exp(t) \left(\frac{1}{6}m_6(t) + \frac{5}{6}m_s(t) \right) \\ m_6(t) = \exp(t) \left(\frac{1}{6}m_6(t) + \frac{1}{6}m_{62}(t) + \frac{4}{6}m_s(t) \right) \\ m_{62}(t) = \exp(t) \left(\frac{1}{6} \cdot 1 + \frac{1}{6}m_6(t) + \frac{4}{6}m_s(t) \right) \end{cases}$$

2. (a) Let's denote by x all available information,

$$x = \begin{pmatrix} y_{100} \\ y_{99} \\ y_{98} \\ u_{99} \end{pmatrix}$$

Let's use $t = 100$:

$$y_{100} = 1 + 0.5y_{98} + u_{100} + u_{99}$$

Using all available information we obtain $u_{100} = 1.5$ and hence

$$y_{101} \mid x \sim \mathcal{N}(1 + 0.5y_{99} + u_{100}; 4)$$

- (b) Here we work with true betas:

$$\mathbb{E}(y_{101} \mid y_{100}) = \mu_y + \frac{\text{Cov}(y_{100}, y_{101})}{\text{Var}(y_{100})}(y_{100} - \mu_y)$$

3. (a) Moment generating function

$$m_N(t) = \sum_{j=0} \exp(tj)(1-h)^j h = h \sum_{j=0} (\exp(t)(1-h))^j = \frac{h}{1 - \exp(t)(1-h)}$$

- (b) As $S = N_1 + N_2 + \dots + N_k$:

$$m_S(t) = \left(\frac{h}{1 - \exp(t)(1-h)} \right)^k$$

- (c) Due to my mistake the limit is easy, 0.

In my dream it was $k \rightarrow \infty$, $k \cdot (1-h) \rightarrow 0.5$ and that would be fun!

4. (a) Let's use Ito's lemma

$$dX_t = f'(t) \cos(2021W_t)dt - 2021f(t) \sin(2021W_t)dW_t + \frac{1}{2}2021^2 f(t) \cos(2021W_t)dt$$

- (b) To make X_t a martingale we should kill dt term.

- (c) As X_t is martingale $\mathbb{E}(X_t) = \mathbb{E}(X_0) = f(0)$. So $\mathbb{E}(\cos(2021W_t)) = f(0)/f(t)$.